This article was downloaded by: [60.29.153.49] On: 25 September 2024, At: 01:59
Publisher: Institute for Operations Research and the Management Sciences (INFORMS)
INFORMS is located in Maryland, USA

Operations Research

Publication details, including instructions for authors and subscription information:
http://pubsonline.informs.org

Finite-Sample Guarantees for Wasserstein Distributionally Robust Optimization: Breaking the Curse of Dimensionality

Rui Gao

To cite this article:
Rui Gao (2023) Finite-Sample Guarantees for Wasserstein Distributionally Robust Optimization: Breaking the Curse of Dimensionality. Operations Research 71(6):2291-2306. https://doi.org/10.1287/opre.2022.2326

Full terms and conditions of use: https://pubsonline.informs.org/Publications/Librarians-Portal/PubsOnLine-Terms-and-Conditions

This article may be used only for the purposes of research, teaching, and/or private study. Commercial use or systematic downloading (by robots or other automatic processes) is prohibited without explicit Publisher approval, unless otherwise noted. For more information, contact permissions@informs.org.

The Publisher does not warrant or guarantee the article's accuracy, completeness, merchantability, fitness for a particular purpose, or non-infringement. Descriptions of, or references to, products or publications, or inclusion of an advertisement in this article, neither constitutes nor implies a guarantee, endorsement, or support of claims made of that product, publication, or service.

Copyright © 2022, INFORMS

Please scroll down for article—it is on subsequent pages

With 12,500 members from nearly 90 countries, INFORMS is the largest international association of operations research (O.R.) and analytics professionals and students. INFORMS provides unique networking and learning opportunities for individual professionals, and organizations of all types and sizes, to better understand and use O.R. and analytics tools and methods to transform strategic visions and achieve better outcomes.
For more information on INFORMS, its publications, membership, or meetings visit http://www.informs.org

https://pubsonline.informs.org/journal/opre

OPERATIONS RESEARCH
Vol. 71, No. 6, November–December 2023, pp. 2291–2306
ISSN 0030-364X (print), ISSN 1526-5463 (online)

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

Methods

Finite-Sample Guarantees for Wasserstein Distributionally Robust Optimization: Breaking the Curse of Dimensionality

Rui Gao[a]

[a] Department of Information, Risk and Operations Management, University of Texas at Austin, Austin, Texas 78712
Contact: rui.gao@mccombs.utexas.edu, https://orcid.org/0000-0003-0145-8577 (RG)

Received: October 6, 2020
Revised: January 4, 2022; April 27, 2022
Accepted: May 7, 2022
Published Online in Articles in Advance: July 20, 2022

Area of Review: Optimization

https://doi.org/10.1287/opre.2022.2326

Copyright: © 2022 INFORMS

Abstract. Wasserstein distributionally robust optimization (DRO) aims to find robust and generalizable solutions by hedging against data perturbations in Wasserstein distance. Despite its recent empirical success in operations research and machine learning, existing performance guarantees for generic loss functions are either overly conservative because of the curse of dimensionality or plausible only in large sample asymptotics. In this paper, we develop a nonasymptotic framework for analyzing the out-of-sample performance for Wasserstein robust learning and the generalization bound for its related Lipschitz and gradient regularization problems. To the best of our knowledge, this gives the first finite-sample guarantee for generic Wasserstein DRO problems without suffering from the curse of dimensionality. Our results highlight that Wasserstein DRO, with a properly chosen radius, balances between the empirical mean of the loss and the variation of the loss, measured by the Lipschitz norm or the gradient norm of the loss. Our analysis is based on two novel methodological developments that are of independent interest: (1) a new concentration inequality controlling the decay rate of large deviation probabilities by the variation of the loss and (2) a localized Rademacher complexity theory based on the variation of the loss.

Supplemental Material: The online appendix is available at https://doi.org/10.1287/opre.2022.2326.

Keywords: distributionally robust optimization • Wasserstein metric • variation regularization • generalization bound • transportation-information inequality

1. Introduction

Distributionally robust optimization (DRO) is an emerging paradigm for statistical learning and decision making under uncertainty. It aims to provide robust and generalizable solutions by hedging against a set of distributions in the minimax sense. Different choices of distributional uncertainty set have been investigated thoroughly (Scarf 1958, Žáčková 1966, Shapiro and Kleywegt 2002, Calafiore and El Ghaoui 2006, Erdoğan and Iyengar 2006, Popescu 2007, Delage and Ye 2010, Goh and Sim 2010, Ben-Tal et al. 2013, Wiesemann et al. 2014, Wozabal 2014, Bayraksan and Love 2015, Jiang and Guan 2016, Wang et al. 2016, Esfahani and Kuhn 2017, Zhao and Guan 2018). In this paper, we focus on Wasserstein DRO (Wozabal 2014, Esfahani and Kuhn 2017, Zhao and Guan 2018, Blanchet and Murthy 2019, Xie 2021, Chen et al. 2022, Gao and Kleywegt 2022):

$$\inf_{\theta\in\Theta} \sup_{\mathbb{P}:\, \mathcal{W}_p(\mathbb{P},\mathbb{P}_n)\le\rho_n} \mathbb{E}_{z\sim\mathbb{P}}[f_\theta(z)],$$

which finds a solution θ from a space Θ to minimize the Wasserstein robust loss, defined as the worst-case expectation of the loss function f_θ among a ball of distributions whose p-Wasserstein distance $\mathcal{W}_p$ to the empirical distribution $\mathbb{P}_n$ of sample size n is at most $\rho_n > 0$. Because of its ability to hedge against data perturbations in high dimensions (Blanchet and Murthy 2019, Gao and Kleywegt 2022) and its regularization effect (Shafieezadeh-Abadeh et al. 2015, 2019; Gao and Kleywegt 2017; Volpi et al. 2018; Blanchet et al. 2019a; Bartl et al. 2021), Wasserstein DRO has recently been studied in many areas in machine learning (Shafieezadeh-Abadeh et al. 2015, Chen and Paschalidis 2018, Gao et al. 2018, Sinha et al. 2018, Volpi et al. 2018, Lee and Raginsky 2018, Blanchet et al. 2019a, Abdullah et al. 2019, Najafi et al. 2019, Smirnova et al. 2019, Blanchet and Kang 2021, Duchi et al. 2020, Levine and Feizi 2020, Derman and Mannor 2020); as well as other fields, such as automatic control (Yang 2017, Shafieezadeh-Abadeh et al. 2018, Cherukuri and Cortés 2019, Yang 2020), finance (Blanchet et al. 2022), energy systems (Duan et al. 2018; Wang et al. 2018a, b), statistics (Shafieezadeh-Abadeh et al. 2015; Gao et al. 2018; Blanchet et al. 2019b; Luo and Mehrotra 2019; Nguyen et al. 2021a, b), and transportation (Carlsson et al. 2018). We refer to Kuhn et al. (2019) for a recent survey.

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

Among Wasserstein distances of different orders, $p = 1, 2$ are of particular interest both practically and theoretically. 1-Wasserstein DRO is useful when the loss function is bounded or has linear growth, and often leads to linear programming reformulation when 1-norm or ∞-norm is used (Shafieezadeh-Abadeh et al. 2015, 2019; Kuhn et al. 2019). 2-Wasserstein DRO applies to a larger class of loss functions such as quadratic loss (Shafieezadeh-Abadeh et al. 2018, Cherukuri and Cortés 2019, Nguyen et al. 2021). Efficient gradient-descent algorithms have been developed by virtue of the convex quadratic subproblem associated with 2-Wasserstein DRO (Li and Montúfar 2018, Sinha et al. 2018, Chen and Li 2020, Blanchet et al. 2022b). Moreover, Lipschitz regularization and data-dependent gradient regularization deep learning problems are closely related to 1-Wasserstein DRO and 2-Wasserstein DRO, respectively.

Like many other (distributionally) robust optimization frameworks or regularization methods, obtaining a Wasserstein robust solution with good performance guarantees requires a proper hyperparameter tuning, namely, the selection of the radius of the Wasserstein ball ρ_n. On the one hand, the radius ρ_n cannot be too small because otherwise, the problem behaves like empirical risk minimization or sample average approximation, thus losing the purpose of robustification. On the other hand, the radius ρ_n cannot be so large that the solution might be overly conservative, which is one of the major criticisms faced by traditional robust optimization. Practically, radius selection is often achieved via cross-validation. From a statistical point of view, it is crucial to understand what is the correct scaling of the hyperparameter ρ_n with respect to the sample size n to ensure the robustness and generalization of the solution without sacrificing much out-of-sample performance.

Despite promising applications of Wasserstein DRO, its theoretical performance guarantee is limited. Esfahani and Kuhn (2017) provides the first out-of-sample performance guarantee for Wasserstein DRO. Using the concentration of empirical Wasserstein distance (Fournier and Guillin 2015), they show that if the radius is chosen in the order of $n^{-1/\max(2,d)}$, where d is the dimension of the random data z, the underlying data-generating distribution $\mathbb{P}_{\text{true}}$ is contained in the Wasserstein ball with high probability. Thereby the Wasserstein robust loss of every feasible solution (and in particular the optimal solution) would be an upper bound of its true loss. This provides a finite-sample nonasymptotic guarantee for the Wasserstein robust solution, but unfortunately, such a bound suffers from the *curse of dimensionality* because the radius shrinks too slow even for problems in moderate dimensions.

To resolve the curse of dimensionality, a series of work by Blanchet et al. (Blanchet et al. 2019a, 2022a; Blanchet and Kang 2021) considers an approach inspired from the empirical likelihood (Lam 2019, Duchi et al. 2021). Their principle is finding the smallest radius ρ_n such that with high probability, the Wasserstein ball contains at least one distribution $\mathbb{P}$, not necessarily equal to the true data-generating distribution $\mathbb{P}_{\text{true}}$, for which there exists an optimal solution to $\min_\theta \mathbb{E}_{\mathbb{P}}[f_\theta]$ that is also optimal to the underlying true problem $\min_\theta \mathbb{E}_{\mathbb{P}_{\text{true}}}[f_\theta]$. Such choice leads to a confidence region of the optimal solution. Through an asymptotic analysis, they show that the radius ρ_n can be chosen in the square-root order $1/\sqrt{n}$ for fixed dimension of the random variable (in Section 4.4 they also derive a result for high-dimensional LASSO problem). This gives the first radius selection rule that does not suffer from the curse of dimensionality. However, one potential issue with this result is that the bound is valid only in the *asymptotic* sense, namely, as the sample size n goes to infinity while fixing the dimension of the random variable. Ideally, we would like to have a performance guarantee for any finite sample size and dimension, especially for high-dimensional problems and for robust optimization where the sample size is comparatively not large.

For certain special classes of stochastic optimization problems, the nonasymptotic $1/\sqrt{n}$-rate has been developed. For 1-Wasserstein DRO with certain linear structure, such as linear regression/classification and their kernelization, Shafieezadeh-Abadeh et al. (2019) shows that the radius can be chosen as $\tilde{O}(1/\sqrt{n})$ to achieve a finite-sample performance guarantee uniformly for all feasible solutions (we use $\tilde{O}$ to suppress the logarithmic dependence). Chen and Paschalidis (2018) derives generalization bounds for certain class of 1-Wasserstein DRO problems that are equivalent to norm regularization. Xie et al. (2021) provides performance guarantees for stochastic bottleneck problems by relating them to sample average approximations.

Yet, it remains largely unknown whether the nonasymptotic $1/\sqrt{n}$-rate holds for general loss functions. In this paper, we provide an affirmative answer to this open question under reasonable assumptions. Informally, our main result states the following performance guarantees for Wasserstein DRO.

Theorem (Informal). *Let* $p \in [1,2]$. *Set* $\rho_n = \tilde{O}(1/\sqrt{n})$. *Under appropriate conditions, with high probability, simultaneously for all* $\theta \in \Theta$,

$$\mathbb{E}_{z\sim\mathbb{P}_{\text{true}}}[f_\theta(z)] \le \sup_{\mathbb{P}:\mathcal{W}_p(\mathbb{P},\mathbb{P}_n)\le\rho_n} \mathbb{E}_{z\sim\mathbb{P}}[f_\theta(z)] + \epsilon_n,$$

where ϵ_n *is a* $\tilde{O}(1/n)$ *higher-order term.*

The constants hidden in the big-O notation are made explicitly in our formal results and are shown to have a mild dependence on the dimension of the random variable for a variety of applications. This theorem

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

shows that under the canonical root-n radius, up to a high-order residual, the true loss is upper bounded by the Wasserstein robust loss uniformly for all θ and particularly for the robust optimal solution θ_{rob}:

$$\mathbb{E}_{z\sim\mathbb{P}_{\text{true}}}[f_{\theta_{\text{rob}}}(z)] \le \sup_{\mathbb{P}:\mathcal{W}_p(\mathbb{P},\mathbb{P}_n)\le\rho_n} \mathbb{E}_{z\sim\mathbb{P}}[f_{\theta_{\text{rob}}}(z)] + \epsilon_n = \min_{\theta\in\Theta} \sup_{\mathbb{P}:\mathcal{W}_p(\mathbb{P},\mathbb{P}_n)\le\rho_n} \mathbb{E}_{z\sim\mathbb{P}}[f_\theta(z)] + \epsilon_n.$$

When f_θ is the loss function of a supervised learning problem, the left-hand side represents the *generalization error* of the Wasserstein robust solution, and the right-hand side of the previous inequality indicates that the optimal value of the Wasserstein robust loss minimization provides an generalization bound up to a high-order residual.

Recall the regularization effect of Wasserstein DRO (Gao and Kleywegt 2017, Blanchet et al. 2019a, Shafieezadeh-Abadeh et al. 2019):

$$\sup_{\mathbb{P}:\mathcal{W}_p(\mathbb{P},\mathbb{P}_n)\le\rho_n} \mathbb{E}_{z\sim\mathbb{P}}[f_\theta(z)] = \mathbb{E}_{z\sim\mathbb{P}_n}[f_\theta(z)] + \rho_n\cdot\mathcal{V}(f_\theta) + \tilde{O}_p(1/n), \quad \forall\,\theta\in\Theta,$$

where $\mathcal{V}(\cdot)$ represents the *variation of the loss*, measured by the Lipschitz norm $\|f_\theta\|_{\text{Lip}}$ when $p=1$ or the gradient norm $\mathbb{E}_{z\sim\mathbb{P}_n}[\|\nabla_z f_\theta(z)\|^2]^{1/2}$ when $p=2$ (we use O_p for the big O in probability notation). Together with this result, our theorem highlights a principled *bias-variation tradeoff* by properly choosing the radius for Wasserstein DRO, which balances between the empirical loss $\mathbb{E}_{\mathbb{P}_n}[f_{\theta_{\text{rob}}}]$ and the variation of the loss $\mathcal{V}(f_{\theta_{\text{rob}}})$ that controls the *generalization gap*:

$$\mathbb{E}_{z\sim\mathbb{P}_{\text{true}}}[f_{\theta_{\text{rob}}}(z)] - \mathbb{E}_{z\sim\mathbb{P}_n}[f_{\theta_{\text{rob}}}(z)] \le \frac{\rho_0}{\sqrt{n}}\cdot\mathcal{V}(f_{\theta_{\text{rob}}}) + \tilde{O}_p(1/n).$$

Thus, the robust optimal solution θ_{rob} achieves nice generalization capability by biasing toward a solution with small variation.

Variation-based regularization has become increasingly popular for many deep learning problems recently. For example, Lipschitz regularization and gradient regularization have shown superior empirical performance for adversarial learning and reinforcement learning (Bousquet et al. 2003, Goodfellow et al. 2015, Lyu et al. 2015, Gulrajani et al. 2017, Sokolić et al. 2017, Finlay et al. 2018, Jakubovitz and Giryes 2018, Miyato et al. 2018, Oberman and Calder 2018, Virmaux and Scaman 2018, Volpi et al. 2018, Terjék 2019, Wei and Ma 2019, Yu et al. 2020). Our results also provide statistical guarantees for Lipschitz regularization and gradient regularization.

We briefly describe two methodological advancements that lead to our results. In our analysis, the main object of study is the *Wasserstein regularizer*:

$$\mathcal{R}_{\mathbb{Q},p}(\rho;f_\theta) = \sup_{\mathbb{P}:\mathcal{W}_p(\mathbb{P},\mathbb{Q})\le\rho} \mathbb{E}_{z\sim\mathbb{P}}[f_\theta(z)] - \mathbb{E}_{z\sim\mathbb{Q}}[f_\theta(z)],$$

that is, the difference between the Wasserstein robust loss and the nominal loss under some distribution $\mathbb{Q}$ such as $\mathbb{P}_n$ or $\mathbb{P}_{\text{true}}$.

First, in Section 3, leveraging tools from transportation-information inequalities (Gozlan and Léonard 2010) in modern probability theory, we derive a new large-deviation type concentration inequality for the empirical loss (Theorem 1). It shows that under proper conditions on the underlying data-generating distribution $\mathbb{P}_{\text{true}}$, the decay rate of the tail probability is upper bounded by the inverse of Wasserstein regularizer $\mathcal{R}^{-1}_{\mathbb{P}_{\text{true}},p}(\cdot;f_\theta)$, as well as the variation of the loss $\mathcal{V}(f_\theta)$. This result shows that the variation of the loss has a direct control on the deviation of the empirical loss from the ground truth. This is an analog of variance-based control often resulting from Chebyshev's or Bernstein's concentration inequalities.

Second, to extend the concentration result above from a single loss function to a family of loss functions, we develop two sets of results in Sections 4.1 and 4.2, respectively, one based on covering number arguments, and the other adapts tools from localized Rademacher complexity theory (Bartlett et al. 2005, Duchi and Namkoong 2019). For the latter, we consider subsets of function classes whose variations are controlled, as opposed to usual approaches based on the mean or variance of the loss. These results are demonstrated in Section 5 using various examples, including feature-based newsvendor, linear prediction, portfolio optimization, Lipschitz regularization for kernel classes, and gradient regularization for neural networks.

Overall, we develop a nonasymptotic statistical analysis framework for Wasserstein DRO and its associated variation regularization and demonstrate the bias-variation tradeoff in Wasserstein robust learning when the radius is properly chosen. This can be served as a counterpart of the well-known bias-variance tradeoff theory in machine learning.

1.1. Related Work

The generalization bounds for robust optimization dates back to Xu and Mannor (2012), which studies generalization of learning algorithms from the viewpoint of robustness. In the introduction, we have elaborated on the literature that provide performance guarantees for Wasserstein DRO (Esfahani and Kuhn 2017; Chen and Paschalidis 2018; Shafieezadeh-Abadeh et al. 2019; Blanchet et al. 2019b, 2021b; Xie et al. 2021; Blanchet and Kang 2021) and discuss their scopes and limitations. In addition to this literature, motivated by distribution shift in domain adaptation and adversarial learning, Lee and Raginsky (2018), Sinha et al. (2018), and Najafi et al. (2019) develop generalization bounds for Wasserstein DRO where the

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

radius is fixed, not varying with the sample size. For divergence DRO and the related variance regularization, Lam (2019) studies the calibration of the radius of divergence ball that recovers the best statistical guarantee. Asymptotics and nonasymptotics of divergence DRO and its bias-variance tradeoff are investigated in Duchi et al. (2021) and Duchi and Namkoong (2019). Besides DRO, Wasserstein distance and transportation-information inequality are also exploited to improve information-theoretic generalization bounds for learning algorithms (Xu and Raginsky 2017; Lopez and Jog 2018; Russo and Zou 2019; Wang et al. 2019, 2021).

The rest of the paper proceeds as follows. In Section 2, we briefly review some results in Wasserstein DRO and its variation regularization effect. We develop a new variation-based concentration inequality in Section 3. Based on these two sections, we derive generalization bounds for variation regularization and the finite-sample guarantees for Wasserstein DRO in Section 4. Applications of these results are in Section 5. We conclude the paper in Section 6. All proofs are deferred to the online appendices.

2. Wasserstein DRO and Variation Regularization

In this section, we introduce notations and provide some background on Wasserstein DRO and its variation regularization effect.

Notation. Let $\mathcal{Z}$ be a Banach space equipped with some norm $\|\cdot\|$ and let $\|\cdot\|_*$ be its dual norm. Define the diameter of $\mathcal{Z}$ as $\operatorname{diam}(\mathcal{Z}) := \sup_{\tilde{z},z\in\mathcal{Z}}\|\tilde{z}-z\|$. Let $p \in [1,\infty)$ and denote by q its Hölder conjugate number, that is, $\frac{1}{p}+\frac{1}{q}=1$. We denote by $\mathcal{P}_p(\mathcal{Z})$ the set of Borel probability measures on $\mathcal{Z}$ with finite pth moment, namely, $\mathbb{Q}\in\mathcal{P}_p(\mathcal{Z})$ if and only if its expectation $\mathbb{E}_{z\sim\mathbb{Q}}[\|z\|^p]<\infty$. Whenever it is clear from the context, we will write $\mathbb{E}_{z\sim\mathbb{P}}$ as $\mathbb{E}_{\mathbb{P}}$ and omit the random variable inside the expectation. The support of a distribution is denoted by $\operatorname{supp}\mathbb{Q}$. The $\mathcal{L}^p(\mathbb{Q})$-norm of a $\mathbb{Q}$-measurable function h is denoted by $\|h\|_{\mathbb{Q},p}=\mathbb{E}_{z\sim\mathbb{Q}}[|h(z)|^p]^{1/p}$. The sup-norm of a function h is denoted by $\|h\|_\infty$, and the Lipschitz norm of a Lipschitz continuous function h is denoted by $\|h\|_{\mathrm{Lip}}$. We denote $a\vee b=\max(a,b)$ and $a\wedge b=\min(a,b)$. For the expectation operator $\mathbb{E}_{z\sim\mathbb{Q}}[\cdot]$, we often write it as $\mathbb{E}_{\mathbb{Q}}[\cdot]$ provided that the involved random variable is clear from the context.

The *Wasserstein distance* of order p between distributions $\mathbb{P},\mathbb{Q}\in\mathcal{P}_p(\mathcal{Z})$ is defined via

$$\mathcal{W}_p(\mathbb{P},\mathbb{Q})^p := \inf_{\pi\in\mathcal{P}_p(\mathcal{Z}^2)}\{\mathbb{E}_{(\tilde{z},z)\sim\pi}[\|\tilde{z}-z\|^p] : \pi \text{ has marginal distributions } \mathbb{P},\mathbb{Q}\}.$$

We denote by $\mathcal{F}:=\{f_\theta:\theta\in\Theta\}$ the class of loss functions. To ease notations, we often suppress the subscript θ and use f to represent a generic loss function from $\mathcal{F}$. Given a nominal distribution $\mathbb{Q}\in\mathcal{P}_p(\mathcal{Z})$ and a radius $\rho\ge 0$, Wasserstein DRO problem is given by

$$\inf_{f\in\mathcal{F}}\sup_{\mathbb{P}\in\mathcal{P}_p(\mathcal{Z})}\{\mathbb{E}_{z\sim\mathbb{P}}[f(z)] : \mathcal{W}_p(\mathbb{P},\mathbb{Q})\le\rho\}.$$

Suppose there exists $M,L\ge 0$ such that $f(z)\le M+L\|z\|^p$ for all $z\in\mathcal{Z}$, then the previous inner maximization problem has a dual problem that always has a minimizer Gao and Kleywegt (2022):

$$\min_{\lambda\ge 0}\left\{\lambda\rho^p+\mathbb{E}_{z\sim\mathbb{Q}}\left[\sup_{\tilde{z}\in\mathcal{Z}}\{f(\tilde{z})-\lambda\|\tilde{z}-z\|^p\}\right]\right\}. \tag{1}$$

In a data-driven problem, the nominal distribution is often chosen as the empirical distribution $\mathbb{P}_n=\frac{1}{n}\sum_{i=1}^n\boldsymbol{\delta}_{z_i^n}$ constructed from n independently and identically distributed (i.i.d.) samples $\{z_i^n\}_{i=1}^n$ from the underlying true distribution $\mathbb{P}_{\mathrm{true}}$, where $\boldsymbol{\delta}_z$ denotes the Dirac point mass on z. We use $\mathbb{P}_\otimes$ or $\mathbb{E}_\otimes$ to indicate that the probability or expectation is evaluated with respect to the sampling distribution, namely the n-fold product distribution $\otimes_{i=1}^n\mathbb{P}_{\mathrm{true}}$ over $\mathcal{Z}^n$.

We define the *Wasserstein regularizer* as the difference between the Wasserstein robust loss and the nominal loss:

$$\mathcal{R}_{\mathbb{Q},p}(\rho;f) := \sup_{\mathbb{P}\in\mathcal{P}_p(\mathcal{Z})}\{\mathbb{E}_{\mathbb{P}}[f] : \mathcal{W}_p(\mathbb{P},\mathbb{Q})\le\rho\}-\mathbb{E}_{\mathbb{Q}}[f].$$

The connection between Wasserstein DRO and regularization has been established under various settings (Shafieezadeh-Abadeh et al. 2015, Esfahani and Kuhn 2017, Gao and Kleywegt 2017, Volpi et al. 2018, Blanchet et al. 2019a, Shafieezadeh-Abadeh et al. 2019, Bartl et al. 2021). The next two results adapted from Gao and Kleywegt (2017) (see also Bartl et al. 2021) establish connections between the Wasserstein regularizer $\mathcal{R}_{\mathbb{Q},p}$ and Lipschitz regularization ($p=1$) and gradient regularization ($p=2$), respectively. For completeness we provide their proofs in Online Appendix EC.1.

Assumption 1. *Assume the following holds:*

(I) *There exists $\gamma_1>0$ such that for every $f\in\mathcal{F}$,*

$$f(\tilde{z})-f(z)\le\gamma_1\|\tilde{z}-z\|,\quad\forall z,\tilde{z}\in\mathcal{Z}.$$

(II) *Suppose* $\operatorname{diam}(\mathcal{Z}):=\sup_{\tilde{z},z\in\mathcal{Z}}\|\tilde{z}-z\|=\infty$, *and for every $f\in\mathcal{F}$, there exists $z_0\in\mathcal{Z}$ such that*

$$\limsup_{\|z-z_0\|\to\infty}\frac{f(z)-f(z_0)}{\|z-z_0\|}=\|f\|_{\mathrm{Lip}}.$$

Assumption 1(I) means that every f is Lipschitz continuous, and Assumption 1(II) means that the Lipschitz norm is attained at infinity.

Lemma 1 (Lipschitz Regularization). *Let $\mathbb{Q}\in\mathcal{P}_1(\mathcal{Z})$ and $\rho\ge 0$. Assume Assumption 1(I) holds, then*

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

$$\mathcal{R}_{\mathbb{Q},1}(\rho;f) \le \rho \cdot \|f\|_{\mathrm{Lip}}.$$

Assume, in addition, Assumption 1(II) holds. Then

$$\mathcal{R}_{\mathbb{Q},1}(\rho;f) = \rho \cdot \|f\|_{\mathrm{Lip}}.$$

Assumption 2. *Assume every $f \in \mathcal{F}$ is differentiable and there exist $\hbar > 0$ such that*

$$\|\nabla f(\tilde{z}) - \nabla f(z)\|_* \le \hbar\|\tilde{z} - z\|, \quad \forall \tilde{z}, z \in \mathcal{Z}, \ \forall f \in \mathcal{F}.$$

This is a smoothness condition that requires that every f has Lipschitz gradient.

Lemma 2 (Gradient Regularization). *Let $\mathbb{Q} \in \mathcal{P}_2(\mathcal{Z})$ and $\rho \ge 0$. Assume Assumption 2 holds. Then*

$$|\mathcal{R}_{\mathbb{Q},2}(\rho;f) - \rho \cdot \|\|\nabla f\|_*\|_{\mathbb{Q},2}| \le \hbar\rho^2.$$

3. Variation-Based Concentration Inequality

In this section, we derive a large-deviation type concentration inequality for the empirical mean of a single loss function. We derive an equivalent representation of $\mathcal{R}^{-1}_{\mathbb{P}_{\mathrm{true}},p}(\cdot;f)$ in Section 3.1 and provide a brief overview of transportation-information inequalities in Section 3.2. The new concentration inequality is developed in Section 3.3, whose proof is postponed to Section EC.2.1.

3.1. Inverse of the Wasserstein Regularizer

Fixing $f \in \mathcal{F}$, we define a function $\mathcal{I}_p : \mathbb{R}_+ \to \mathbb{R}_+ \cup \{+\infty\}$ via

$$\mathcal{I}_p(\varepsilon;f)^p := \sup_{t>0}\left\{\varepsilon t - \mathbb{E}_{z\sim\mathbb{P}_{\mathrm{true}}}\left[\sup_{\tilde{z}\in\mathcal{Z}}\{t(f(\tilde{z}) - f(z)) - \|\tilde{z} - z\|^p\}\right]\right\},$$

which will play a similar role as the rate function in the large deviation principle. The next proposition establishes its connection to the Wasserstein regularizer $\mathcal{R}_{\mathbb{P}_{\mathrm{true}},p}$, whose proof is given in Online Appendix EC.2.

Proposition 1. *Let $p \in [1,\infty)$ and $f \in \mathcal{F}$. Suppose there exists $M, L \ge 0$ such that $f(z) \le M + L\|z\|^p$ for all $z \in \mathcal{Z}$. Let $\rho > 0$ and suppose the dual minimizer λ_o of (1) is positive. Set $\underline{\lambda} := \lim_{\|z\|\to\infty} f(z)/\|z\|^p$. Then*

$$\mathcal{I}_p(\mathcal{R}_{\mathbb{P}_{\mathrm{true}},p}(\rho;f);f)\begin{cases} = \rho, \text{if } \ \lambda_o > \underline{\lambda}, \\ \ge \rho, \text{if } \ \lambda_o = \underline{\lambda}. \end{cases}$$

The dual optimizer of (1) tends to be large when ρ is close to zero, in which case $\lambda_o > \underline{\lambda}$, as observed in Gao and Kleywegt (2022). Hence Proposition 1 shows that at least for small ρ, the left inverse of $\mathcal{R}_{\mathbb{P}_{\mathrm{true}},p}(\cdot;f)$ is precisely $\mathcal{I}_p(\cdot;f)$.

3.2. Transportation-Information Inequalities

Just like many other results on the concentration of measure, appropriate conditions on the function f and the distribution $\mathbb{P}_{\mathrm{true}}$ are required. Because we are dealing with general loss functions that are possibly unbounded, some assumptions on the underlying data-generating distribution are necessary. It turns out for our purpose, it is convenient to work with the *transportation-information inequality*, a useful condition to establish concentration of measure in modern probability theory.

Definition 1 (Transportation-Information Inequality). Let $p \in [1,\infty)$. A distribution $\mathbb{P} \in \mathcal{P}_p(\mathcal{Z})$ satisfies a *transportation-information inequality* $T_p(\tau)$ for some positive constant τ, if

$$\mathcal{W}_p(\mathbb{Q},\mathbb{P}) \le \sqrt{\tau H(\mathbb{Q}\|\mathbb{P})}, \ \ \forall \mathbb{Q} \in \mathcal{P}_p(\mathcal{Z}),$$

where $H(\mathbb{Q}\|\mathbb{P})$ denotes the relative entropy $H(\mathbb{Q}\|\mathbb{P}) := \int_{\mathcal{Z}} \log(d\mathbb{Q}/d\mathbb{P})\,d\mathbb{Q}$, where $d\mathbb{Q}/d\mathbb{P}$ denotes the Radon-Nikodym derivative.

We briefly comment on distributions satisfying transportation-information inequalities and refer the reader to Gozlan and Léonard (2010) for a recent survey and (Villani 2008, chapter 22) for an in-depth discussion. Among different choices of p, T_1 and T_2 are of particular interest and have been widely studied in the literature; T_1 is equivalent to the following condition (theorem 2.3 in Djellout et al. 2004, theorem 22.10 in Villani 2008; see also Lemma EC.1 in Online Appendix EC.2): a distribution $\mathbb{P}$ satisfies T_1 if and only if there exists $a > 0$ such that $\mathbb{E}[\exp(a\|z\|^2)] < \infty$. In particular, any distribution on a bounded support $\mathcal{Z}$ with $\mathrm{diam}(\mathcal{Z}) < \infty$ satisfies $T_1(2\mathrm{diam}(\mathcal{Z})^2)$. T_2 is also known as Talagrand's inequality, which is implied by the log-Sobolev inequality. Examples of distributions satisfying the log-Sobolev inequality include distributions with a strongly log-concave density, and mixture of distributions satisfying log-Sobolev inequality whose pairwise chi-squared divergences are uniformly bounded (Chen et al. 2021). For $p_1 \le p_2$, T_{p_1} is weaker than T_{p_2} because $\mathcal{W}_{p_1} \le \mathcal{W}_{p_2}$. In the sequel, we will focus on the case $p \in [1,2]$.

3.3. Concentration for a Single Loss Function

Now we are ready to state our main result in this section.

Theorem 1 (Variation-Based Concentration). *Let $p \in [1,2]$ and $f \in \mathcal{F}$. Assume there exist $M, L > 0$ such that*

$$f(z) \le M + L\|z\|^p, \quad \forall z \in \mathcal{Z}.$$

Assume further that $\mathbb{P}_{\mathrm{true}}$ satisfies $T_p(\tau)$ for some $\tau > 0$. Let $\varepsilon > 0$. Then

$$\mathbb{P}_{\otimes}\{\mathbb{E}_{\mathbb{P}_n}[f] - \mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[f] < -\varepsilon\} \le \exp(-n\mathcal{I}_p(\varepsilon;-f)^2/\tau).$$

Let $t > 0$. Then with probability at least $1 - e^{-t}$,

$$\mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[f] \ \le \ \mathbb{E}_{\mathbb{P}_n}[f] + \mathcal{R}_{\mathbb{P}_{\mathrm{true}},p}\left(\sqrt{\frac{\tau t}{n}};-f\right). \tag{2}$$

Theorem 1 uncovers an interesting connection: the nonasymptotic decay rate of large deviation probabilities is

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

controlled by the inverse of Wasserstein regularizer $\mathcal{I}_p(\varepsilon;-f)$. The negative sign $-f$ appears because here we bound the downside risk, that is, the probability of empirical loss being smaller than the true loss, whereas $\mathcal{R}_{\mathbb{P}_{\text{true}},p}$ is defined via upside excess, that is, the worst-case loss that is greater than the true loss. As a matter of fact, a similar result holds if we swap the empirical loss and true loss in the theorem:

$$\mathbb{P}_{\otimes}\{\mathbb{E}_{\mathbb{P}_n}[f]-\mathbb{E}_{\mathbb{P}_{\text{true}}}[f]>\varepsilon\}\le\exp(-n\mathcal{I}_p(\varepsilon;f)^2/\tau),$$

and with probability at least $1-e^{-t}$,

$$\mathbb{E}_{\mathbb{P}_n}[f]\le\mathbb{E}_{\mathbb{P}_{\text{true}}}[f]+\mathcal{R}_{\mathbb{P}_{\text{true}},p}\left(\sqrt{\frac{\tau t}{n}};f\right).$$

When $p=1$, if f is Lipschitz continuous, then by Lemma 1, we have

$$\mathcal{R}_{\mathbb{P}_{\text{true}},p}\left(\sqrt{\frac{\tau t}{n}};-f\right)\le\sqrt{\frac{\tau t}{n}}\cdot\|-f\|_{\text{Lip}}=\sqrt{\frac{\tau t}{n}}\cdot\|f\|_{\text{Lip}}.$$

When $p=2$, if f has Lipschitz continuous gradient, then by Lemma 2 we have

$$\mathcal{R}_{\mathbb{P}_{\text{true}},2}\left(\sqrt{\frac{\tau t}{n}};-f\right)\le\sqrt{\frac{\tau t}{n}}\cdot\|\|\nabla f\|_*\|_{\mathbb{P}_{\text{true}},2}+\frac{\hbar\tau t}{n}.$$

Substituting these inequalities in Theorem 1 yields following corollary.

Corollary 1 (Variation Regularization). *Let $p\in\{1,2\}$. When $p=1$, assume Assumption 1(I) holds; when $p=2$, assume Assumption 2 holds. Assume further that $\mathbb{P}_{\text{true}}$ satisfies $\mathsf{T}_p(\tau)$ for some $\tau>0$. Let $t>0$. Then with probability at least $1-e^{-t}$,*

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f]\le\mathbb{E}_{\mathbb{P}_n}[f]+\begin{cases}\sqrt{\frac{\tau t}{n}}\cdot\|f\|_{\text{Lip}}, & p=1,\\ \sqrt{\frac{\tau t}{n}}\cdot\|\|\nabla f\|_*\|_{\mathbb{P}_{\text{true}},2}+\frac{\hbar\tau t}{n}, & p=2.\end{cases}$$

Theorem 1 and Corollary 1 show that the Wasserstein regularizer $\mathcal{R}_{\mathbb{P}_{\text{true}},p}\left(\sqrt{\frac{\tau t}{n}};-f\right)$, as well as the variation of the loss, $\|f\|_{\text{Lip}}$ or $\|\|\nabla f\|_*\|_{\mathbb{P}_{\text{true}},2}$, are natural quantities controlling the deviation of the empirical loss for distributions satisfying a transportation-information inequality. For $p=1$, thanks to the first part of Lemma 1, the bound in Theorem 1 is tighter than the Lipschitz norm bound in Corollary 1, which was obtained in Bobkov and Götze (1999). Because $\|\|\nabla f\|_*\|_{\mathbb{P}_{\text{true}},2}\le\|f\|_{\text{Lip}}$, $p=2$ suggests a tighter upper bound than $p=1$, at the cost of a stronger assumption on the underlying distribution.

4. Finite-Sample Guarantees

In the previous section, we have derived a concentration inequality for a single loss function, and the goal of this section is to extend it to a family of loss functions $\mathcal{F}$. In the spirit of Esfahani and Kuhn (2017), Blanchet et al. (2019b), and Shafieezadeh-Abadeh et al. (2019), we would like to determine a proper scaling of the Wasserstein radius ρ_n with respect to sample size n so that with high probability, the Wasserstein robust loss is an upper bound of the true loss uniformly for all functions in the class $\mathcal{F}=\{f_\theta:\theta\in\Theta\}$. Whenever this holds, minimizing the Wasserstein robust loss controls the true loss as well.

When $\mathcal{F}$ is a finite set, then a simple application of the union bound to Theorem 1 yields that (2) holds simultaneously for all $f\in\mathcal{F}$ with probability at least $1-|\mathcal{F}|e^{-t}$, where $|\cdot|$ denotes the cardinality of a set. When $\mathcal{F}$ contains infinitely many functions, some notion of complexity of the function class $\mathcal{F}$ is needed to obtain uniform convergence. In Section 4.1, we prove the result using a standard covering number argument; and in Section 4.2, we adopt techniques from local Rademacher complexity theory (Bartlett et al. 2005, Koltchinskii 2006).

4.1 Covering Number Arguments

Recall that for $\epsilon>0$, the *covering number* $\mathcal{N}(\epsilon;\mathcal{H},\|\cdot\|_{\mathcal{H}})$ of a set $\mathcal{H}$ with respect to a norm $\|\cdot\|_{\mathcal{H}}$ is defined as the smallest cardinality of an ϵ-cover of $\mathcal{H}$, where $\mathcal{H}_\epsilon$ is an ϵ-cover of $\mathcal{H}$ if for each $h\in\mathcal{H}$, there exists $\tilde{h}\in\mathcal{H}_\epsilon$ such that $\|\tilde{h}-h\|_{\mathcal{H}}\le\epsilon$. Similar to the classic stochastic programming literature (Shapiro et al. 2014, section 5.3.2), we can obtain a union bound using the standard covering number argument, whose proof is given in Onlien Appendix EC.3.1. Throughout this section, we let $\mathcal{F}=\{f_\theta:\theta\in\Theta\}$ and we impose the following smoothness assumption with respect to the parameter θ.

Assumption 3. *Assume there exists a measurable function $\kappa:\mathcal{Z}\to\mathbb{R}_+$ and constants $\kappa_M,\kappa_L\ge0$ satisfying $\kappa(z)\le\kappa_M+\kappa_L\|z\|^p$ for all $z\in\mathcal{Z}$, such that*

$$|f_{\tilde{\theta}}(z)-f_\theta(z)|\le\kappa(z)\|\tilde{\theta}-\theta\|_\Theta,$$
$$\forall\theta,\tilde{\theta}\in\Theta,\ \mathbb{P}_{\text{true}}-a.e.\ z\in\mathcal{Z}.$$

Corollary 2. *Let $p\in[1,2]$. Assume $\mathbb{P}_{\text{true}}$ satisfies $\mathsf{T}_p(\tau)$ for some $\tau>0$ and Assumption 3 holds.*

Let $t>0$. Set

$$\rho_n=\sqrt{\frac{\tau(t+\log\mathcal{N}(1/n;\Theta,\|\cdot\|_\Theta))}{n}},$$

and $\epsilon_n=(2\mathbb{E}_{\mathbb{P}_{\text{true}}}[\kappa]+\sqrt{\mathbb{V}\text{ar}_{\mathbb{P}_{\text{true}}}[\kappa]}+\kappa_M+2^{p-1}\kappa_L((\tau t/n)^{p/2}+\mathbb{E}_{\mathbb{P}_{\text{true}}}[\|z\|^p]))/n$. Then we have the following:

(I) *Assume there exist constants $M,L>0$ such that*

$$f_\theta(z)\le M+L\|z\|^p,\quad\forall z\in\mathcal{Z},\ \forall\theta\in\Theta.$$

Then with probability at least $1-1/n-e^{-t}$,

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f_\theta]\le\mathbb{E}_{\mathbb{P}_n}[f_\theta]+\mathcal{R}_{\mathbb{P}_{\text{true}},p}(\rho_n;-f_\theta)+\epsilon_n,\quad\forall\theta\in\Theta.$$

(II) *Lipschitz Regularization and* 1-Wasserstein DRO: *When $p=1$, assume Assumption 1(I) holds. Then with*

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

probability at least $1-1/n-e^{-t}$,

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f_\theta] \le \mathbb{E}_{\mathbb{P}_n}[f_\theta] + \rho_n \cdot \|f_\theta\|_{\text{Lip}} + \epsilon_n, \quad \forall \theta \in \Theta.$$

Assume, in addition, that Assumption 1(II) *holds. Then with probability at least* $1-1/n-e^{-t}$,

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f_\theta] \le \mathbb{E}_{\mathbb{P}_n}[f_\theta] + \mathcal{R}_{\mathbb{P}_n,1}(\rho_n; f_\theta) + \epsilon_n, \quad \forall \theta \in \Theta.$$

(III) Gradient regularization: *When* $p=2$, *assume Assumption* 2 *holds. Then with probability at least* $1-1/n-e^{-t}$,

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f_\theta] \le \mathbb{E}_{\mathbb{P}_n}[f_\theta] + \rho_n \cdot \| \|\nabla f_\theta\|_*\|_{\mathbb{P}_{\text{true}},2} + \hbar\rho_n^2 + \epsilon_n, \ \forall \theta \in \Theta.$$

Corollary 2 establishes the generalization bounds for Wasserstein DRO and the Lipschitz and gradient regularization. By Wainwright 2019 (example 5.8), $\log \mathcal{N}(\epsilon; \Theta, \|\cdot\|_*) \le d \log(1+2B/\epsilon)$, where B is the diameter of Θ. Thereby, by choosing the radius $\rho_n = \tilde{O}(\sqrt{d/n})$, the Wasserstein robust loss serves as an upper bound of the true loss for all $f_\theta \in \mathcal{F}$ up to an $O(1/n)$ remainder for $p=1$ and an $O(d/n)$ remainder for $p=2$. Assumption 1(II) may be restrictive for certain applications. In Example 1, we demonstrate an instance for which finite-sample guarantee holds with an-$O(1/\sqrt{n})$ radius even though this assumption does not hold. Essentially, as long as the Wasserstein regularizer can be sandwiched by multiples of Lipschitz regularizer (see theorem 2 in Gao and Kleywegt 2017), similar finite-sample guarantees can be obtained; we refer to a follow-up work (An and Gao 2021) for more in-depth discussions.

In the next result, we provide an empirical counterpart of Corollary 2 for $p=2$.

Assumption 4. *Assume there exists a measurable function* $\kappa_2 : \mathcal{Z} \to \mathbb{R}_+$ *and constants* $\kappa_{2,M}, \kappa_{2,L} \ge 0$ *satisfying* $\kappa_2(z) \le \kappa_{2,M} + \kappa_{2,L}\|z\|^p$ *for all* $z \in \mathcal{Z}$, *such that*

$$\|\nabla f_{\tilde{\theta}}(z) - \nabla f_\theta(z)\|_* \le \kappa_2(z)\|\tilde{\theta} - \theta\|_\Theta, \quad \forall \theta, \tilde{\theta} \in \Theta, \ \mathbb{P}_{\text{true}} - a.e.\ z \in \mathcal{Z}.$$

Corollary 3 (2-Wasserstein DRO). *Assume* $\mathbb{P}_{\text{true}}$ *satisfies* $T_2(\tau)$ *for some* $\tau > 0$ *and Assumptions* 2–4 *hold. Assume* $\sigma = \sup_{\theta\in\Theta} \mathbb{E}_{\mathbb{P}_{\text{true}}}[\|\nabla f_\theta\|_*^4]^{\frac{1}{2}} / \| \|\nabla f_\theta\|_*\|^2_{\mathbb{P}_{\text{true}},2} < \infty$. *Let* $t > 0$ *and* $n > 8\sigma^2 t$. *Set*

$$\rho_n = \sqrt{\frac{\tau(t + \log(1+\mathcal{N}(1/n; \Theta, \|\cdot\|_\Theta)))}{n}} \left(1 + \sigma\sqrt{\frac{2(t+\log(1+\mathcal{N}(1/n;\Theta,\|\cdot\|_\Theta)))}{n}}\right),$$

and

$$\tilde{\epsilon}_n = (2\mathbb{E}_{\mathbb{P}_{\text{true}}}[\kappa_2] + \sqrt{\mathbb{V}\text{ar}_{\mathbb{P}_{\text{true}}}[\kappa_2]} + \rho_n\sqrt{\mathbb{E}_{\mathbb{P}_{\text{true}}}[\kappa_2^2] + \sqrt{\mathbb{V}\text{ar}_{\mathbb{P}_{\text{true}}}[\kappa_2^2]}})/n.$$

Then with probability at least $1-2/n-2e^{-t}$,

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f_\theta] \le \mathbb{E}_{\mathbb{P}_n}[f_\theta] + \rho_n \| \|\nabla f_\theta\|_*\|_{\mathbb{P}_n,2} + \tilde{\epsilon}_n + \frac{\hbar\tau(t+\log(1+\mathcal{N}(1/n;\Theta,\|\cdot\|_\Theta)))}{n}, \quad \forall \theta \in \Theta,$$

and

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f_\theta] \le \mathbb{E}_{\mathbb{P}_n}[f_\theta] + \mathcal{R}_{\mathbb{P}_n,2}(\rho_n; f_\theta) + \tilde{\epsilon}_n + \frac{2\hbar\tau(t+\log(1+\mathcal{N}(1/n;\Theta,\|\cdot\|_\Theta)))}{n}, \quad \forall \theta \in \Theta.$$

In Section 5, we will demonstrate Corollary 2 for feature-based newsvendor problem in Example 1 and Corollary 3 for linear prediction with Lipschitz loss in Example 3.

4.2. Local Rademacher Complexity Arguments

The covering number bound developed in the previous section may be loose. Indeed, the discussion after Corollary 2 indicates that for smooth parametric family the radius $\tilde{O}(\sqrt{d/n})$ is always dimension dependent. To obtain a tighter bound in a more general setting, we derive results using local Rademacher complexity theory. As it turns out, this approach sometimes leads to a bound that has a better dependence on d and even dimension-independent bound (see Examples 2, 4, and 5).

Let us begin with some technical preparation. Recall the *Rademacher complexity* of a function class $\mathcal{F}$ with respect to a sample $\{z_i^n\}_{i=1}^n$ is defined as

$$\mathfrak{R}_n(\mathcal{F}) := \mathbb{E}_{\boldsymbol{\sigma}}\left[\sup_{f\in\mathcal{F}} \frac{1}{n}\sum_{i=1}^n \boldsymbol{\sigma}_i f(z_i^n)\right],$$

where $\boldsymbol{\sigma}_i$ s are i.i.d. Rademacher random variables with $\mathbb{P}\{\boldsymbol{\sigma}_i = \pm 1\} = \frac{1}{2}$. The Rademacher complexity of the function class $\mathcal{F}$ with respect to $\mathbb{P}_{\text{true}}$ for sample size n is defined as $\mathbb{E}_\otimes[\mathfrak{R}_n(\mathcal{F})]$. Rademacher complexity plays an important role in bounding the generalization error of statistical learning problems but may be vacuous if $\mathcal{F}$ is too large. The idea of localization is to restrict on a small subset of $\mathcal{F}$ around the optimal solution that often admits low complexity. The *localized Rademacher complexity* (Bartlett et al. 2005) at level $r > 0$ is defined as

$$\mathbb{E}_\otimes[\mathfrak{R}_n(\{cf : f \in \mathcal{F}, 0 \le c \le 1, T(cf) \le r\})],$$

where $T : \mathcal{F} \to \mathbb{R}_+$. In our analysis, we choose $T(f) = \|f\|^2_{\text{Lip}}$ when $p=1$ and $T(f) = \| \|\nabla f\|_*\|^2_{\mathbb{P}_{\text{true}},2}$ when $p=2$. Part of our techniques below are adapted from the framework developed in Bartlett et al. (2005) and Duchi and Namkoong (2019), which primarily considers $T(f) = \mathbb{E}_{\mathbb{P}_{\text{true}}}[f^2]$.

By choosing a proper level r_n, the localized Rademacher complexity of the functions of the subset can be much smaller than the entire family, which enables a better bound. Often, the level r_n is chosen to be the fixed point r_{n*} of some function $\psi_n(r)$, which serves as

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

an upper bound on the localized Rademacher complexity at level r. A typical assumption imposed on ψ_n is the so-called subroot condition. A function $\psi: \mathbb{R}_+ \to \mathbb{R}_+$ is *subroot* if it is nonconstant, nonnegative, nondecreasing and the map $r \longmapsto \psi(r)/\sqrt{r}$ is nonincreasing for all $r > 0$. A subroot function always has a unique fixed point $r_{n\star}$ (Bartlett et al. 2005). Similar to the literature, we impose the following assumption.

Assumption 5 (Subroot Local Complexity). *Assume there exists a subroot function $\psi_n : \mathbb{R}_+ \to \mathbb{R}_+$ such that*

$$\psi_n(r) \geq \mathbb{E}_{\otimes}[\mathfrak{R}_n(\{cf : f \in \mathcal{F}, 0 \leq c \leq 1, T(cf) \leq r\})].$$

Denote by $r_{n\star}$ the fixed point of ψ_n.

We will verify this assumption for various examples considered in Section 5.

We first study the case of $p = 1$. The proof is given in Online Appendix EC.3.2.2.

Theorem 2 (Lipschitz Regularization). *Assume $\mathbb{P}_{\text{true}}$ satisfies $T_1(\tau)$, Assumption 1(I) holds, and Assumption 5 holds with $T(f) = \|f\|_{\text{Lip}}^2$. Let $t > 0$. Then with probability at least $1 - \lceil \log_2(\sqrt{\gamma_1 \tau t n}) \rceil e^{-t}$,*

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f] \leq \mathbb{E}_{\mathbb{P}_n}[f] + \left(2\sqrt{\frac{\tau t}{n}} + \sqrt{4r_{n\star} + \frac{2}{n}}\right)\|f\|_{\text{Lip}} + 4r_{n\star} + \frac{2}{n}, \quad \forall f \in \mathcal{F}.$$

Together with Lemma 1, we obtain the following result.

Corollary 4 (1-Wasserstein DRO). *Assume $\mathbb{P}_{\text{true}}$ satisfies $T_1(\tau)$, Assumption 1(I) holds and Assumption 5 holds with $T(f) = \|f\|_{\text{Lip}}^2$. Let $t > 0$. Set*

$$\rho_n = 2\sqrt{\frac{\tau t}{n}} + \sqrt{4r_{n\star} + \frac{2}{n}}.$$

Then with probability at least $1 - \lceil \log_2(\sqrt{\gamma_1 \tau t n}) \rceil e^{-t}$,

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f] \leq \mathbb{E}_{\mathbb{P}_n}[f] + \mathcal{R}_{\mathbb{P}_n,1}(\rho_n; f) + 4r_{n\star} + \frac{2}{n}, \quad \forall f \in \mathcal{F}.$$

Note that $\rho_n = \tilde{O}(1/\sqrt{n})$ if $r_{n\star} = \tilde{O}(\frac{1}{n})$. A sufficient condition for this to hold is the subroot function $\psi_n(r) = \tilde{O}(\sqrt{r/n})$, which holds for many important cases as we illustrated in Section 5. As such, Theorem 2 and Corollary 4 show that by choosing a radius in the order of $1/\sqrt{n}$, with high probability, the Wasserstein robust loss serves as a upper bound for the true loss up to an $\tilde{O}(1/n)$ gap. Here the probability bound has a $O(\log n)$ term, nearly independent of sample size n. By mapping t to $t + \log\lceil \log_2(\sqrt{\gamma_1 \tau t n}) \rceil$, one can obtain a probability bound that is independent of sample size, whereas the radius $\rho_n \sim O(\sqrt{\log \log n / n})$. In the rest of the paper, we will not make such a transformation, but just keep in mind that these two results are equivalent.

In the next corollary, we consider the loss functions of a composition form $\ell \circ f$, where $\ell : \mathbb{R} \to \mathbb{R}$ is a given Lipschitz function and $f \in \mathcal{F}$, which occurs often in supervised learning. The following result is useful to establish the generalization bound for problems of this type.

Corollary 5 (Lipschitz Composition). *Assume $\mathbb{P}_{\text{true}}$ satisfies $T_1(\tau)$, Assumption 1(I) holds, and Assumption 5 holds with $T(f) = \|f\|_{\text{Lip}}^2$. Let ℓ be an L_ℓ-Lipschitz function and $t > 0$. Then with probability at least $1 - \lceil \log_2(\sqrt{L_\ell \gamma_1 \tau t n}) \rceil e^{-t}$,*

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[\ell \circ f] \leq \mathbb{E}_{\mathbb{P}_n}[\ell \circ f] + 2\left(\sqrt{\frac{\tau t}{n}} L_\ell + \sqrt{4L_\ell^2 r_{n\star} + \frac{2L_\ell}{n}}\right)\|f\|_{\text{Lip}} + 4L_\ell^2 r_{n\star} + \frac{2L_\ell}{n}, \quad \forall f \in \mathcal{F}.$$

In Section 5, we will illustrate Corollary 5 in supervised learning with linear class (Example 2) and with nonlinear kernel class (Example 5).

The analysis for 2-Wasserstein DRO is aligned with the previous case but requires more care to deal with the data-dependent regularization $\| \|\nabla f\|_* \|_{\mathbb{P}_n,2}$; see details in Online Appendix EC.3.2.3. Define the family of normalized gradient norm functions

$$\mathcal{G} := \left\{ \frac{\|\nabla f\|_*^2}{\| \|\nabla f\|_* \|_{\mathbb{P}_{\text{true}},2}^2} : f \in \mathcal{F} \right\}.$$

Theorem 3 (Gradient Regularization). *Assume that $\mathbb{P}_{\text{true}}$ satisfies $T_2(\tau)$, Assumption 2 holds, and Assumption 5 holds with $T(f) = \| \|\nabla f\|_* \|_{\mathbb{P}_{\text{true}},2}^2$. Assume there exists $\gamma_2 > 0$ such that $\| \|\nabla f\|_* \|_{\mathbb{P}_{\text{true}},2} \leq \gamma_2$ for all $f \in \mathcal{F}$. Let $t > 0$. Set*

$$\rho_n = 2\sqrt{\frac{\tau t}{n}}(1 + \mathbb{E}_{\otimes}[\mathfrak{R}_n(\mathcal{G})]) + \sqrt{4r_{n\star} + 2\epsilon_n},$$

and

$$\epsilon_n = \frac{\hbar \tau t + 1 + \mathbb{E}_{\otimes}[\mathfrak{R}_n(\mathcal{G})]}{n}.$$

Then with probability at least $1 - \lceil \log_2(\sqrt{\gamma_2 \tau t n}) \rceil e^{-t}$,

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f] \leq \mathbb{E}_{\mathbb{P}_n}[f] + \rho_n \| \|\nabla f\|_* \|_{\mathbb{P}_{\text{true}},2} + 4r_{n\star} + 2\epsilon_n, \quad \forall f \in \mathcal{F}.$$

Whenever $\mathbb{E}_{\otimes}[\mathfrak{R}_n(\mathcal{G})] = O(1)$ and $r_{n\star} = O(1/n)$, Theorem 3 shows that by choosing a radius in the order of $1/\sqrt{n}$, with high probability, the gradient regularized loss serves as an upper bound for the true loss up to an $O(1/n)$ gap. Note that $\mathbb{E}_{\otimes}[\mathfrak{R}_n(\mathcal{G})] = O(1/\sqrt{n})$ as long as $\mathcal{G}$ has finite VC dimension (lemma 4.14 and proposition 4.18 in Wainwright 2019). In Section 5, we will illustrate this result in portfolio optimization (Example 4) and neural networks (Example 6) that show that $\mathbb{E}_{\otimes}[\mathfrak{R}_n(\mathcal{G})] = O(1/\sqrt{n})$ and $r_{n\star} = O(1/n)$ with explicit constants.

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

The distribution-dependent gradient regularization $\| \|\nabla f\|_*\|_{\mathbb{P}_{\text{true}},2}$ in Theorem 3 can be replaced with its empirical counterpart $\| \|\nabla f\|_*\|_{\mathbb{P}_n,2}$. In the next result, we provide the generalization bound for data-dependent gradient regularization problems and 2-Wasserstein DRO.

Corollary 6 (2-Wasserstein DRO). *Under the setting in Theorem 3, assume additionally that there exists $\kappa_g > 0$ such that $\frac{\|\nabla f(z)\|_*}{\| \|\nabla f\|_*\|_{\mathbb{P}_{\text{true}},2}} \le \kappa_g$ for all $f \in \mathcal{F}$ and $z \in \mathcal{Z}$. Set*

$$\tilde{\rho}_n = \rho_n\left(1 + 2\mathbb{E}_{\otimes}[\mathfrak{R}_n(\mathcal{G})] + \kappa_g^2\sqrt{\frac{t}{2n}}\right).$$

Then whenever $2\mathbb{E}_{\otimes}[\mathfrak{R}_n(\mathcal{G})] + \kappa_g^2\sqrt{\frac{t}{2n}} < 1/2$, with probability at least $1 - (\lceil \log_2(\sqrt{\gamma_2 \tau t n})\rceil + 1)e^{-t}$, for every $f \in \mathcal{F}$,

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f] \le \mathbb{E}_{\mathbb{P}_n}[f] + \tilde{\rho}_n \| \|\nabla f\|_*\|_{\mathbb{P}_n,2} + 4r_{n\star} + 2\epsilon_n,$$

and

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f] \le \mathbb{E}_{\mathbb{P}_n}[f] + \mathcal{R}_{\mathbb{P}_n,2}(\tilde{\rho}_n; f) + 4r_{n\star} + 2\epsilon_n + \hbar\tilde{\rho}_n^2.$$

Remark 1 (Performance Bound on the Robust Optimal Solution). Denote $\mathcal{L}(\theta) := \mathbb{E}_{\mathbb{P}_{\text{true}}}[f_\theta]$, $\mathcal{L}_n(\theta) := \mathbb{E}_{\mathbb{P}_n}[f_\theta]$ and $\mathcal{L}_n^{\text{rob}}(\theta;\rho) := \sup_{\mathbb{P}:\mathcal{W}_p(\mathbb{P},\mathbb{P}_n)\le\rho} \mathbb{E}_{\mathbb{P}}[f_\theta]$. Then the high-probability bounds we have developed in this section have the form

$$\mathcal{L}(\theta) \le \mathcal{L}_n^{\text{rob}}(\theta;\rho_n) + \tilde{\epsilon}_n, \quad \forall \theta \in \Theta.$$

Let $\theta_* \in \arg\min_{\theta\in\Theta}\mathcal{L}(\theta)$. Using the decomposition

$$\begin{aligned}
&\mathcal{L}(\theta_{\text{rob}}) - \mathcal{L}(\theta_*) \\
&\quad = (\mathcal{L}(\theta_{\text{rob}}) - \mathcal{L}_n^{\text{rob}}(\theta_{\text{rob}};\rho_n)) + (\mathcal{L}_n^{\text{rob}}(\theta_{\text{rob}};\rho_n) \\
&\qquad - \mathcal{L}_n^{\text{rob}}(\theta_*;\rho_n)) + (\mathcal{L}_n^{\text{rob}}(\theta_*;\rho_n) - \mathcal{L}_n(\theta_*)) + (\mathcal{L}_n(\theta_*) \\
&\qquad - \mathcal{L}(\theta_*)) \\
&\quad \le \tilde{\epsilon}_n + 0 + (\rho_n\mathcal{V}(\theta_*) + O(\rho_n^2)) + (\mathcal{L}_n(\theta_*) - \mathcal{L}(\theta_*)).
\end{aligned}$$

Observe that $\mathcal{L}_n(\theta_*) - \mathcal{L}(\theta_*) = O_p(1/\sqrt{n})$ under mild conditions. Hence, when $\tilde{\epsilon}_n = \tilde{O}(1/n)$ and $\rho_n = \tilde{O}(1/\sqrt{n})$, the right-hand side provides an $\tilde{O}(1/\sqrt{n})$ bound on the losses between the robust optimal solution and the true optimal solution. If $\mathcal{L}$ satisfies the Polyak-Łojasiewicz condition (or equivalently, the quadratic growth condition) (Karimi et al. 2016):

$$\mathcal{L}(\theta) - \mathcal{L}(\theta_*) \ge \frac{\mu}{2}\|\theta - \theta_*\|^2, \quad \forall \theta \in \Theta,$$

where $\mu > 0$, then $\|\theta_{\text{rob}} - \theta_*\|$ is upper bounded by a multiple of $\mathcal{L}(\theta_{\text{rob}}) - \mathcal{L}(\theta_*)$, thereby $\|\theta_{\text{rob}} - \theta_*\| = \tilde{O}_p(1/\sqrt{n})$.

5. Applications

In this section, we demonstrate our theoretical results in the context of various applications in operations research and machine learning.

5.1. Performance Guarantees for Wasserstein DRO

5.1.1. Big-Data Newsvendor. We first consider a big-data newsvendor problem in the spirit of Ban and Rudin (2019). In this problem, the decision maker needs to find the optimal ordering quantity for a product with an unknown random demand y, subject to holding cost $h > 0$ and back-order cost $b > 0$. In the world of big data, before deciding the ordering quantity, the decision maker observes a vector of features (such as product information, customer profiles, economic indicators) and thus can use them make a better ordering decision using these feature information. The vector of features is modeled as a d-dimensional random variable X and the decision maker has collected historical observations of the feature-demand vector of the product. The goal is to find a decision rule that maps every realization x from the feature space $\mathcal{X} \subset \mathbb{R}^d$ to an ordering decision. For illustration, we focus on a simple linear decision rule parameterized by $\theta \in \mathbb{R}^d$, thereby the ordering quantity for a product with feature $x \in \mathcal{X}$ is $\theta^\top x$. Thus, for a given θ, the expected cost equals $\mathbb{E}_{\mathbb{P}}[h(\theta^\top x - y)_+ + b(y - \theta^\top x)_+]$, where $\mathbb{P}$ is the joint distribution of feature-demand vector $z = (x, y)$.

Example 1 (Feature-Based Newsvendor). Let $\mathcal{Z} = (\mathcal{X} \times \mathbb{R}_+, \|\cdot\|)$ be the space of feature-demand vectors. Consider the following distributionally robust feature-based newsvendor problem:

$$\min_{\theta\in\Theta} \sup_{\mathbb{P}:\mathcal{W}_1(\mathbb{P},\mathbb{P}_n)\le\rho_n} \mathbb{E}_{\mathbb{P}}[h(\theta^\top x - y)_+ + b(y - \theta^\top x)_+].$$

Suppose $\Theta \subset \{\theta \in \mathbb{R}^d : \|\theta\|_* \le B\}$ and $\mathbb{P}_{\text{true}}$ satisfies $T_1(\tau)$. Let $f_\theta(x,y) = h(\theta^\top x - y)_+ + b(y - \theta^\top x)_+$ and $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$.

We have $|f_{\tilde{\theta}}(z) - f_\theta(z)| \le (h \vee b)\|x\| \|\tilde{\theta} - \theta\|_*$ and thus Assumption 3 holds with $\kappa(z) = (h \vee b)\|x\|$. Hence, by Corollary 2(II), setting $\epsilon_n = \frac{1}{n}\left(3\mathbb{E}_{\mathbb{P}_{\text{true}}}[\|x\|] + \mathbb{V}\text{ar}_{\mathbb{P}_{\text{true}}}[\|x\|] + \sqrt{\frac{\tau t}{n}}\right) = O(1/n)$, with probability at least $1 - 1/n - \exp(-t)$, we have

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f_\theta] \le \mathbb{E}_{\mathbb{P}_n}[f_\theta] + \sqrt{\frac{\tau t(1 + d\log(1 + 2Bn))}{n}}\|f_\theta\|_{\text{Lip}} + \epsilon_n, \quad \forall \theta \in \Theta.$$

Set $\rho_n = \frac{h\vee b}{h\wedge b}\sqrt{\frac{\tau t(1+d\log(1+2Bn))}{n}}$. By Lemma EC.13 in Online Appendix EC.4.1, we have $\rho_n\|f_\theta\|_{\text{Lip}} \le \mathcal{R}_{\mathbb{P}_n,1}(f_\theta;\rho_n) = \tilde{O}(\sqrt{d/n})$. Thus, with probability at least $1 - 1/n - \exp(-t)$,

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[f_\theta] \le \mathbb{E}_{\mathbb{P}_n}[f_\theta] + \mathcal{R}_{\mathbb{P}_n,1}(f_\theta;\rho_n) + \epsilon_n, \quad \forall \theta \in \Theta. \quad \clubsuit$$

We remark that in this example, the newsvendor loss function satisfies only Assumption 1(I) but not Assumption 1(II). Hence, we did not directly use

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

Corollary 2(II) to derive the performance guarantee. Instead, Lemma EC.13 in Online Appendix EC.4.1 actually shows that $\mathcal{R}_{\mathbb{P}_n,1}(f_\theta;\rho_n)$ can achieve a fraction $\frac{h\wedge b}{h\vee b}$ of $\|f_\theta\|_{\mathrm{Lip}}$ (and thus $\mathcal{R}_{\mathbb{P}_{\mathrm{true}},1}(f_\theta;\rho_n)$) uniformly for all θ, thereby we can still choose $\rho_n = \tilde{O}(\sqrt{d/n})$ to ensure a good performance guarantee for the Wasserstein robust solution.

5.1.2. Linear Prediction. We consider supervised learning with linear models. Let $z=(x,y)\in\mathcal{Z}=\mathcal{X}\times\mathcal{Y}\subset\mathbb{R}^d\times\mathbb{R}$. To ease the exposition, we assume $\|z-\tilde{z}\| = \|x-\tilde{x}\|_2 + \infty\mathbf{1}\{y\neq\tilde{y}\}$, thereby we only focus on the x-component when calculating the gradient. Set

$$l(u,y) := \begin{cases} \ell(u-y), & \text{regression,} \\ \ell(yu), & \text{classification,} \end{cases} \tag{3}$$

where $\ell:\mathbb{R}\to\mathbb{R}$ is L-Lipschitz continuous, and $\mathcal{Y}\subset\mathbb{R}$ for regression and $\mathcal{Y}=\{\pm 1\}$ for classification, and denote $l\circ f_\theta(z) := l(f_\theta(x),y)$. We denote by l' the derivative function of l with respect to its first argument, which is well defined almost everywhere in $\mathbb{R}$. Denote by $\mathbb{P}^x_{\mathrm{true}}$ the x-marginal distribution of $\mathbb{P}_{\mathrm{true}}$. The two examples considered in this subsection are on linear predictions for $p=1$ and $p=2$, respectively.

Example 2 (Linear Class with Lipschitz Loss, 1-Wasserstein DRO). Let $\Theta\subset\{\theta\in\mathbb{R}^d : \|\theta\|_2\le B\}$ for some $B>0$. Define

$$\mathcal{F} = \{x\longmapsto f_\theta(x)=\theta^\top x : \theta\in\Theta\}.$$

Consider loss functions of the composite form (3) and let $f_\theta(x)=\theta^\top x$. Assume $\mathbb{P}^x_{\mathrm{true}}$ is sub-Gaussian, that is, there exists $a>0$ such that $C := \log\mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[\exp(a\|x\|_2^2)] < \infty$. Assume ℓ satisfies $\limsup_{|t|\to\infty}\frac{\ell(t)}{|t|}=L$. Examples of $\ell(t)$ include convex losses such as hinge loss $(1-t)_+$, softplus (logistic) loss $\log(1+e^t)$, as well as nonconvex losses such as inverse S-shaped curve $\mathrm{sgn}(t)\log(\frac{1}{2}(1+e^t))$.

Let us verify the assumptions in Corollary 5. By Lemma EC.1 in Online Appendix EC.2, $\mathbb{P}^x_{\mathrm{true}}$ satisfies $T_1(\frac{2}{a}(1+C))$. Assumption 1 holds with $\|f_\theta\|_{\mathrm{Lip}} = \|\theta\|_2 \le B = \gamma_1$. Furthermore, because

$$\{cf_\theta : \theta\in\Theta, 0\le c\le 1, c^2\|\theta\|_2^2\le r\}\subset\{f_\theta : \|\theta\|_2\le\sqrt{r}\},$$

by Lemma EC.14 in Online Appendix EC.4.2, $\sqrt{r\mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[\|x\|_2^2]/n}$ is an upper bound of $\mathbb{E}_\otimes[\mathfrak{R}_n(\{f_\theta : \|\theta\|_2\le\sqrt{r}\})]$. By Jensen's inequality, we have

$$\mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[\|x\|_2^2] \le \frac{1}{a}\log\mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[\exp(a\|x\|_2^2)] = \frac{C}{a}.$$

It follows that we can set $\psi_n(r)=\sqrt{\frac{rC}{an}}$ and thus $r_{n\star}=\frac{C}{an}$.

Let $t>0$. Set

$$\rho_n = 2\left(L\sqrt{\frac{2(1+C)t}{an}}+\sqrt{\epsilon_n}\right), \quad \epsilon_n = 4L^2\frac{C}{an}+\frac{2L}{n}.$$

Then $\rho_n = O(1/\sqrt{n})$ and $\epsilon_n = O(1/n)$ are dimension independent, and by Corollary 5 and Lemma 1, with probability at least $1-\lceil(\frac{1}{2}\log_2(LB\frac{2}{a}(1+C)tn)\rceil e^{-t}$,

$$\begin{aligned}\mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[l(\theta^\top x,y)] - \mathbb{E}_{\mathbb{P}_n}[l(\theta^\top x,y)] &\le \mathcal{R}_{\mathbb{P}_n,1}(\rho_n; l\circ f_\theta) \\ &\quad + \epsilon_n = \rho_n\|\theta\|_2+\epsilon_n, \quad \forall\theta\in\Theta. \quad \clubsuit\end{aligned}$$

The bound obtained in Example 2 is consistent to the existing literature on the generalization bounds for linear predictions. However, unlike the typical results (Shalev-Shwartz and Ben-David 2014, Chen and Paschalidis 2018), we do not impose boundedness assumptions on the loss function ℓ or the domain $\mathcal{Z}$. If imposing a positive lower bound on $\|\theta\|_2\ge c>0$, the bound given in the example further becomes

$$\begin{aligned}\mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[l(\theta^\top x,y)] - \mathbb{E}_{\mathbb{P}_n}[l(\theta^\top x,y)] &\le (\rho_n+\epsilon_n/c)\|\theta\|_2 \\ &= \mathcal{R}_{\mathbb{P}_n,1}(\rho_n+\epsilon_n/c; l\circ f_\theta).\end{aligned}$$

Thereby one can bound the true loss using only the Wasserstein robust loss with an inflated radius $\rho_n+\epsilon_n/c$ without having a higher-order error term. This bound is of the same form as in (Shafieezadeh-Abadeh et al. 2019, theorem 39) which has a linear dependence on the dimension d of $\mathcal{X}$ (albeit under a light-tail assumption that is slightly weaker than T_1), whereas our bound $\rho_n = O(1/\sqrt{n})$ is independent of d (at least for the case of 2-norm).

Example 3 (Linear Class with Lipschitz Loss, 2-Wasserstein DRO). Consider a similar setup as in Example 2 but with slightly different notations in order to be consistent with Corollary 3. Let $\Theta\subset\{\theta\in\mathbb{R}^d : \|\theta\|_*\le B\}$ for some $B>0$. Define

$$\mathcal{F} = \{(x,y)\longmapsto l(\theta^\top x,y) : \theta\in\Theta\},$$

where l is defined in (3). Let $f_\theta(z)=l(\theta^\top x,y)$. Then $\|\nabla f_\theta(z)\|_* = \|\nabla_x l(\theta^\top x,y)\|_* = \|\theta\|_*|l'(\theta^\top x,y)|$, recalling l' denotes the derivative of l with respect to its first argument. Assume further that ℓ in (3) has $\hbar$-Lipschitz gradient; $\mathbb{P}^x_{\mathrm{true}}$ satisfies $T_2(\tau)$ for some $\tau>0$; and $\inf_{\theta\in\Theta}\mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[l'(\theta^\top x,y)^2]>0$. The last condition is mild; indeed, it is satisfied if for every $\theta\in\Theta$, $l'(\theta^\top\cdot,y)$ is nonzero on some subset of $\mathcal{X}$ with positive $\mathbb{P}^x_{\mathrm{true}}$-measure (together with the boundedness assumption on Θ).

Let us verify the conditions and compute the constants in Corollary 3. Assumption 2 is satisfied because f_θ has $\hbar B^2$-Lipschitz gradient, and $\sigma = \sup_{\theta\in\Theta}\mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[l'(\theta^\top x,y)^4]^{\frac{1}{2}} / \mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[l'(\theta^\top x,y)^2] \le L^2 / \inf_{\theta\in\Theta}\mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[l'(\theta^\top x,y)^2] < \infty$. We have $|f_{\tilde{\theta}}(z)-f_\theta(z)|\le L_\ell\|x\|\,\|\tilde{\theta}-\theta\|_*$, and by Lemma EC.15, $\|\nabla f_\theta(z)-\nabla f_{\tilde{\theta}}(z)\|_*\le(L_\ell+B\hbar\|x\|)\|\tilde{\theta}-\theta\|_*$. Hence Assumptions 3 and 4 hold.

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

Let $t > 0$ and set

$$\rho_n = \sqrt{\frac{\tau t(1 + d\log(2+2Bn))}{n}} \left(1 + \sigma\sqrt{\frac{2t(1+d\log(2+2Bn))}{n}}\right),$$

$$\tilde{\epsilon}_n = (2L_\ell + 2B\hbar\, \mathbb{E}_{\mathbb{P}_{\text{true}}}[\|x\|] + B^2\hbar^2\, \mathbb{V}\text{ar}_{\mathbb{P}_{\text{true}}}[\|x\|] + \rho_n \sqrt{\mathbb{E}_{\mathbb{P}_{\text{true}}}[(L_\ell + B\hbar\|x\|)^2] + \mathbb{V}\text{ar}_{\mathbb{P}_{\text{true}}}[(L_\ell + B\hbar\|x\|)^2]})/n.$$

By Corollary 3, with probability at least $1 - 2/n - 2\exp(-t)$, for every $\theta \in \Theta$,

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[l(\theta^\top x, y)] - \mathbb{E}_{\mathbb{P}_n}[l(\theta^\top x, y)] \le \mathcal{R}_{\mathbb{P}_n,2}(\rho_n; f_\theta) + \tilde{\epsilon}_n + \frac{2\hbar B^2\tau t(1 + d\log(2+2Bn))}{n}.$$

We have $\rho_n = \tilde{O}(\sqrt{d/n})$ and an $\tilde{O}(d/n)$ remainder.

5.1.3. Portfolio Optimization. In this section, we study the classic Markowitz's mean-variance portfolio optimization problem. Let x be a vector of random losses of d assets with distribution $\mathbb{P}^x_{\text{true}}$ and let $w \in \mathcal{W} \subset \mathbb{R}^d$ be the portfolio weights satisfying $w^\top \mathbf{1} = 1$. The variance of a one-dimensional random variable Y has an equivalent representation $\mathbb{V}\text{ar}[Y] = \min_{u\in\mathbb{R}} \mathbb{E}[(Y-u)^2]$.

Example 4 (Markowitz Model). Let $\alpha > 0$ and suppose $\mathcal{W} \subset \{w \in \mathbb{R}^d : w^\top \mathbf{1} = 1, \|w\|_2 \le B\}$, where $B > 0$. Consider the following distributionally robust mean-variance minimization

$$\min_{w\in\mathcal{W}, u\in\mathbb{R}} \sup_{\mathbb{P}:\mathcal{W}_2(\mathbb{P}^x, \mathbb{P}^x_n)\le\rho_n} \mathbb{E}_{\mathbb{P}^x}[(w^\top x - u)^2 + \alpha w^\top x].$$

Assume $\mathbb{P}^x_{\text{true}}$ satisfies $T_2(\tau)$. Then it also satisfies $T_1(\tau)$ and by Corollary 1, for every $w \in \mathbb{R}^d$ with $\|w\|_2 = 1$, $\mathbb{P}_\otimes\{|\mathbb{E}_{\mathbb{P}^x_n}[w^\top x] - \mathbb{E}_{\mathbb{P}^x_{\text{true}}}[w^\top x]| \ge \epsilon\} \le 2e^{-n\epsilon^2/\tau}$ for all $\epsilon > 0$; thus, $\mathbb{P}^x_{\text{true}}$ is $\sqrt{\tau/2}$-sub-Gaussian. Let $\mu_j = \mathbb{E}_{\mathbb{P}^x_{\text{true}}}[\|x\|_2^j]^{\frac{1}{j}} < \infty$, $j = 1,2,3,4$. Assume there exists $\zeta > 0$ such that $\mathbb{C}\text{ov}_{\mathbb{P}_{\text{true}}}[x] \succeq \zeta I$.

Set $U_n := B(\mu_1 + \sqrt{\tau t/n} + \rho_n)$ and $U := \sup_n U_n$, which is bounded whenever $\{\rho_n\}_n$ is bounded. By Lemma EC.16 in Online Appendix EC.4.3, with probability at least $1 - e^{-t}$, the problem is equivalent to

$$\min_{w\in\mathcal{W}, |u|\le U_n} \sup_{\mathbb{P}:\mathcal{W}_2(\mathbb{P}^x, \mathbb{P}^x_n)\le\rho_n} \mathbb{E}_{\mathbb{P}^x}[(w^\top x - u)^2 + \alpha w^\top x].$$

Let $\theta = (w, \tilde{u})$, where $\tilde{u} = -u + \alpha/2$, $z = (x, y)$ and $f_\theta(z) = (w^\top x + \tilde{u}^\top y)^2 + \alpha u - \alpha^2/4 = (\theta^\top z)^2 - \alpha\tilde{u} + \alpha^2/4$. Thereby we have $f_\theta(x, 1) = (w^\top x - u)^2 + \alpha w^\top x$. Set $\Theta = \{(w, \tilde{u}) : w \in \mathcal{W}, |-\tilde{u} + \alpha/2| \le U\}$. For any distribution $\mathbb{P}^x$, we represent $\mathbb{P} = \mathbb{P}^x \otimes \boldsymbol{\delta}_1$. Consider the problem

$$\min_{\theta=(w,\tilde{u})\in\Theta} \left\{ \sup_{\mathbb{P}:\mathcal{W}_2(\mathbb{P}, \mathbb{P}_n)\le\rho_n} \mathbb{E}_{\mathbb{P}}[(\theta^\top z)^2] - \alpha\tilde{u} + \alpha^2/4 \right\}.$$

Let us verify the assumptions and compute the constants in Theorem 3 for the inner maximization problem: $\mathbb{P}_{\text{true}}$ satisfies $T_2(\tau)$ because any distribution $\mathbb{Q} \in \mathcal{P}_2(\mathcal{Z})$ with finite $H(\mathbb{Q}, \mathbb{P}_{\text{true}})$ has the form $\mathbb{Q}^x \otimes \boldsymbol{\delta}_1$, where $\mathbb{Q}^x$ is the x-marginal of $\mathbb{Q}$. We have $\nabla f_\theta(z) = 2(\theta^\top z)\theta$, thus $\nabla^2 f_\theta(z) = 2\theta\theta^\top$. Note that $\|\theta\|_2^2 \le B^2 + (U + \alpha/2)^2$; hence, Assumption 2 is satisfied with $\hbar = 2(B^2 + (U + \alpha/2)^2)$. To find a subroot function $\psi_n(r)$ required by Assumption 5, observe that

$$\|\,\|\nabla f_\theta\|_2\|_{\mathbb{P}_{\text{true}},2} = 2\|\theta\|_2 \mathbb{E}_{\mathbb{P}_{\text{true}}}[(\theta^\top z)^2]^{\frac{1}{2}} \ge 2\sqrt{\zeta}\|\theta\|_2^2.$$

It follows from Lemma EC.17 in Online Appendix EC.4.3 that

$$\mathbb{E}_\otimes\left[\mathfrak{R}_n\left(\left\{cf_\theta : \theta \in \Theta, 0 \le c \le 1, c^2\|\,\|\nabla f_\theta\|_2\|^2_{\mathbb{P}_{\text{true}},2} \le r\right\}\right)\right] \le \mu_4^2\sqrt{\frac{r}{4\zeta n}} := \psi_n(r).$$

Thus, $r_{n\star} = \frac{\mu_4^4}{4\zeta n}$. Moreover, by Lemma EC.18 in Online Appendix EC.4.3,

$$\mathbb{E}_\otimes\left[\mathfrak{R}_n\left(\left\{\frac{\|\nabla f_\theta\|_2^2}{\|\,\|\nabla f_\theta\|_2\|^2_{\mathbb{P}_{\text{true}},2}} : \theta \in \Theta\right\}\right)\right] \le \frac{1}{1\wedge\zeta}\sqrt{\frac{\mu_4^4 + 2\mu_2^2 + 1}{n}} =: R_n.$$

In addition, $\sup_{\theta\in\Theta}\|\,\|\nabla f_\theta\|_2\|_{\mathbb{P}_{\text{true}},2} = \sup_{\theta\in\Theta} 2\|\theta\|_2 \mathbb{E}_{\mathbb{P}_{\text{true}}}[(\theta^\top z)^2]^{\frac{1}{2}} \le 2\mu_2(B^2 + (U + \alpha/2)^2) = \gamma_2$.

Let $t > 0$. Then in Theorem 3, set

$$\rho_n = 2\sqrt{\frac{\tau t}{n}}(1 + R_n) + \sqrt{\frac{\mu_4^4}{\zeta n} + 2\epsilon_n},$$

$$\epsilon_n = \frac{2(B^2 + (U + \alpha/2)^2)\tau t + 1 + R_n}{n}.$$

We have with probability at least $1 - \lceil\log_2(\sqrt{\gamma_2\tau t n})\rceil e^{-t}$,

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[(\theta^\top z)^2] \le \mathbb{E}_{\mathbb{P}_n}[(\theta^\top z)^2] + \rho_n\|\,\|\nabla f_\theta\|_2\|_{\mathbb{P}_{\text{true}},2} + \epsilon_n, \quad \forall\theta \in \Theta.$$

By Lemma EC.20 in Online Appendix EC.4.3, a generalized version of Corollary 6 applied to unbounded gradient norm, we set

$$\tilde{\rho}_n = C(1 + R_n), \quad \tilde{\epsilon}_n = \epsilon_n + 2(B^2 + (U + \alpha/2)^2)\tilde{\rho}_n^2,$$

where C depends only on the distribution of $\|z\|_2$, $z \sim \mathbb{P}_{\text{true}}$. Then $\tilde{\rho}_n = \tilde{O}(1/\sqrt{n})$, $\tilde{\epsilon}_n = O(1/n)$ are both dimension-independent, and with probability at least $1 - (\lceil\log_2(\sqrt{\gamma_2\tau t n})\rceil + 2)e^{-t}$,

$$\mathbb{E}_{\mathbb{P}_{\text{true}}}[(\theta^\top z)^2] \le \mathbb{E}_{\mathbb{P}_n}[(\theta^\top z)^2] + \mathcal{R}_{\mathbb{P}_n,p}(\tilde{\rho}_n; f_\theta) + \tilde{\epsilon}_n, \ \forall\theta \in \Theta,$$

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

and therefore by minimizing over u or $\tilde{u}$,

$$\mathbb{V}\mathrm{ar}_{\mathbb{P}_{\mathrm{true}}}[w^\top x] + \alpha \mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[w^\top x] \le \min_{u\in\mathbb{R}} \sup_{\mathbb{P}:\mathcal{W}_2(\mathbb{P}^x,\mathbb{P}_n^x)\le\tilde{\rho}_n} \mathbb{E}_{\mathbb{P}^x}[(w^\top x - u)^2 + \alpha w^\top x] + \tilde{\epsilon}_n, \quad \forall w \in \mathcal{W}. \quad \clubsuit$$

The last inequality in Example 4 shows that the true mean-variance of the portfolio w_n is upper bounded by its robust mean-variance up to a higher-order term. We remark that in this example, the parameter space is not bounded as $u \in \mathbb{R}$, which makes it impossible to obtain a bounded complexity for the entire class of loss functions. We circumvent such a difficulty by showing that there exists an optimal solution lying in a bounded set Θ with high probability; thereby, it suffices to restrict on Θ.

5.2. Performance Guarantees for Variation Regularization

In the next two examples, we illustrate our results for Lipschitz regularization and gradient regularization for nonlinear classes. Similar to Section 5.1.2, we let $z = (x, y) \in \mathcal{Z} = \mathcal{X} \times \mathcal{Y} \subset \mathbb{R}^d \times \mathbb{R}$ and assume $\|z - \tilde{z}\| = \|x - \tilde{x}\|_2 + \infty \mathbf{1}\{y \ne \tilde{y}\}$.

5.2.1. Kernel Method. We consider Lipschitz regularization of kernel class (Wainwright 2019, chapter 12). Let $\boldsymbol{k} : \mathcal{X} \times \mathcal{X} \to \mathbb{R}_+$ be a positive definite kernel on $\mathcal{X} \subset (\mathbb{R}^d, \|\cdot\|_2)$ with $\sigma := (\mathbb{E}_{x\sim\mathbb{P}_{\mathrm{true}}}[\boldsymbol{k}(x,x)])^{\frac{1}{2}} < \infty$. We can associate $\boldsymbol{k}$ with a feature map $\Phi : \mathcal{X} \to \mathcal{H}$, where $\mathcal{H}$ is a Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and $\boldsymbol{k}(x_1, x_2) = \langle \Phi(x_1), \Phi(x_2) \rangle$. Denote by $\|\cdot\|_{\mathcal{H}}$ a norm on $\mathcal{H}$. Let $m \in \mathbb{N}_{\ge 1}$ and $\{x_j\}_{j=1}^\infty \subset \mathcal{X}$. Then we have $\|\sum_{j=1}^m \theta_j \Phi(x_j)\|^2 = \sum_{j,k=1}^m \theta_j \theta_k \boldsymbol{k}(x_j, x_k)$.

In kernel method, one often consider the following parameterized class:

$$\mathcal{F} = \left\{ x \longmapsto \sum_{j=1}^\infty \theta_j \boldsymbol{k}(x, x_j) : \theta \in \Theta \right\},$$
$$\text{where } \Theta = \{\theta = (\theta_j)_{j=1}^m : \|\theta\|_{\boldsymbol{k}} \le B, \ m \in \mathbb{N}_{\ge 1}\},$$

for some $B > 0$, where for $\theta = (\theta_j)_{j=1}^m$, we adopt the convention $\sum_{j=1}^\infty \theta_j \boldsymbol{k}(x, x_j) = \sum_{j=1}^m \theta_j \boldsymbol{k}(x, x_j)$ and $\|\theta\|_{\boldsymbol{k}}^2 = \sum_{j,k=1}^\infty \theta_j \theta_k \boldsymbol{k}(x_j, x_k) = \sum_{j,k=1}^m \theta_j \theta_k \boldsymbol{k}(x_j, x_k)$.

Example 5 (Lipschitz Regularization for Kernel Class). Consider loss functions of the form (3) with $u = f_\theta \in \mathcal{F}$ defined previously. Assume $\boldsymbol{k}$ is differentiable and there exists $\zeta > 0$ such that $\| \|\nabla f_\theta\|_2 \|_{\mathbb{P}_{\mathrm{true}},2}^2 = \mathbb{E}_{x\sim\mathbb{P}_{\mathrm{true}}}[\|\sum_{j=1}^\infty \theta_j \nabla_x \boldsymbol{k}(x, x_j)\|_2^2] \ge \zeta \sum_{j,k=1}^\infty \theta_j \theta_k \boldsymbol{k}(x_j, x_k)$ for all $\theta \in \Theta$, which can be satisfied when the matrix $(\mathbb{E}_{x\sim\mathbb{P}_{\mathrm{true}}}[\nabla_x \boldsymbol{k}(x, x_j)^\top \nabla_x \boldsymbol{k}(x, x_k)])_{1\le j,k\le\infty}$ is positive definite. Furthermore, assume $\gamma_1 = \sup_{\theta\in\Theta, x\in\mathcal{X}} \|\sum_{j=1}^\infty \theta_j \nabla_x \boldsymbol{k}(x, x_j)\| < \infty$; thus, Assumption 1(I) is satisfied.

Let us compute the generalization bound using Corollary 5. To this end, we need to specify ψ_n in Assumption 5 and compute its fixed point $r_{n\star}$. Observe that

$$\left\{ c f_\theta : \theta \in \Theta, 0 \le c \le 1, c^2 \|f_\theta\|_{\mathrm{Lip}}^2 \le r \right\} \subset \left\{ c f_\theta : \theta \in \Theta, 0 \le c \le 1, \zeta c^2 \|\theta\|_{\boldsymbol{k}}^2 \le r \right\} \subset \left\{ f_\theta : \|\theta\|_{\boldsymbol{k}} \le \sqrt{r/\zeta} \right\},$$

hence $\psi_n(r)$ can be chosen as $2\sigma\sqrt{\frac{r}{n\zeta}}$, an upper bound of $\mathbb{E}_{\otimes}[\mathfrak{R}_n(\{f_\theta : \|\theta\|_{\boldsymbol{k}} \le \sqrt{r/\zeta}\})]$ according to Bartlett and Mendelson (2002, lemma 22). Thus, $r_{n\star} = \frac{4\sigma^2}{n\zeta}$. Set

$$\rho_n = 2\left(L\sqrt{\frac{\tau t}{n}} + \sqrt{\epsilon_n} \right), \quad \epsilon_n = \frac{16 L^2 \sigma^2}{n\zeta} + \frac{2L}{n}.$$

Then $\rho_n = O(1/\sqrt{n})$ and $\epsilon_n = O(1/n)$ are dimension independent, and by Corollary 5, with probability at least $1 - \lceil \log_2(\sqrt{L\gamma_1 \tau t n}) \rceil e^{-t}$,

$$\mathbb{E}_{\mathbb{P}_{\mathrm{true}}}[l \circ f_\theta] \le \mathbb{E}_{\mathbb{P}_n}[l \circ f_\theta] + \rho_n \|f_\theta\|_{\mathrm{Lip}} + \epsilon_n, \quad \forall \theta \in \Theta. \quad \clubsuit$$

This result provides a generalization bound for Lipschitz regularization problems (Luxburg and Bousquet 2004, Finlay et al. 2018, Virmaux and Scaman 2018) when the loss function class belongs to a kernel class. We remark that the setup in this example is different from (Shafieezadeh-Abadeh et al. 2019, section 3.3), in which the distributional uncertainty is imposed on the feature space, whereas Example 5 considers distributional uncertainty on the original data space.

5.2.2. Neural Networks. In the last example, we illustrate the generalization bound of gradient regularization for a simple two-layer neural network. Consider

$$\mathcal{F} = \{(x, y) \longmapsto l(W_2 \phi(W_1 x), y) : (W_1, W_2) \in \Theta\},$$

where l is defined in (3), $\phi = (\phi_1, \ldots, \phi_{d_2})$ are entry-wise 1-Lipschitz activation functions, and Θ is the space of weight matrices

$$\Theta = \{\theta = (W_1, W_2) : W_1 \in \mathbb{R}^{d_2 \times d_1}, W_2 \in \mathbb{R}^{1 \times d_2}, W_1 W_1^\top = I, \|W_2\|_2 \le B\}.$$

Here the constraint $W_1 W_1^\top = I$ enforces the orthonormal regularization on the weight matrix (Vorontsov et al. 2017, Xie et al. 2017, Bansal et al. 2018, Huang et al. 2018), which is a popular way to ensure the training stability and performance for neural nets. Let $f_\theta(z) = l(W_2 \phi(W_1 x), y)$.

Example 6 (Gradient Regularization for Neural Networks). Assume l and ϕ_j has $\hbar_l$- and $\hbar_\phi$-Lipschitz gradient, respectively, $j = 1, \ldots, d_2$; thereby, f_θ is smooth and $\mathcal{F}$ satisfies Assumption 2 with $\hbar = 2L(L\hbar_\phi + B\hbar_l)\sqrt{d_2}$ by Lemma EC.23 in Online Appendix EC.4.4. Assume

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

$\eta := \inf_{\theta\in\Theta, z\in\mathcal{Z}} l'(W_2\phi(W_1x), y) > 0$, which can be satisfied, for example, when $\mathcal{Z}$ is bounded and l is the logistic loss. Furthermore, assume there exists $\zeta > 0$ such that $\mathbb{E}_{\mathbb{P}^x_{\text{true}}}[\|W_2\phi'(W_1x)\|_2^2] \geq \zeta\|W_2\|_2^2$ for every $(W_1, W_2) \in \Theta$, where $\phi' = (\phi'_1, \ldots, \phi'_{d_2})$, and $\sigma = \mathbb{E}_{\mathbb{P}^x_{\text{true}}}[\|x\|_2^2]^{\frac{1}{2}} < \infty$.

Let us compute the constants in Corollary 6. We have

$$\begin{aligned}\|\nabla f(z)\|_* &= \|l'(W_2\phi(W_1x), y)W_2\phi'(W_1x)W_1\|_2^2 \\ &= l'(W_2\phi(W_1x), y)^2\|W_2\phi'(W_1x)\|_2^2.\end{aligned}$$

Thus, $\|\nabla f(z)\|_* \leq LB$, $\|\,\|\nabla f\|_*\|_{\mathbb{P}_{\text{true}},2} \geq \eta\sqrt{\zeta}\|W_2\|_2$. It follows that

$$\frac{\|\nabla f(z)\|_*}{\|\,\|\nabla f\|_*\|_{\mathbb{P}_{\text{true}},2}} \leq \frac{LB}{\eta\sqrt{\zeta}\|W_2\|_2} =: \kappa_g.$$

In addition, $c^2\|\,\|\nabla f\|_*\|_{\mathbb{P}_{\text{true}},2}^2 \leq r$ implies $\|W_2\|_2 \leq \frac{\sqrt{r}}{c\eta\sqrt{\zeta}}$. As a result, we can choose $\psi_n(r)$ in Assumption 5 as

$$\begin{aligned}&\mathbb{E}_{\otimes}[\mathfrak{R}_n\{cf_\theta : \theta\in\Theta, 0 \leq c \leq 1, c^2\|\,\|\nabla f_\theta\|_*\|_{\mathbb{P}_{\text{true}},2}^2 \leq r\}] \\ &\leq L\mathbb{E}_{\otimes}\left[\mathfrak{R}_n\left\{x \longmapsto W_2\phi(W_1x) : W_1W_1^\top = I, \|W_2\|_2 \leq \frac{\sqrt{r}}{\eta\sqrt{\zeta}}\right\}\right] \\ &\leq \frac{L\sigma\sqrt{2rd_2}}{\eta\sqrt{\zeta n}} =: \psi_n(r),\end{aligned}$$

where the first inequality follows from Lemma EC.6 in Online Appendix EC.4, and the second inequality is from Lemma EC.22 in Online Appendix EC.4.4. It follows that $r_{n\star} = \frac{2L^2\sigma^2d_2}{\eta^2\zeta n}$. Moreover, by Lemma EC.23 in Online Appendix EC.4.4,

$$\mathbb{E}_{\otimes}[\mathfrak{R}_n(\mathcal{G})] \leq \frac{2L(L\hbar_\phi + B\hbar_l)\sigma\sqrt{2d_2}}{\eta^2\zeta\sqrt{n}} =: R_n.$$

Thereby, in Corollary 6, setting

$$\begin{aligned}\tilde{\rho}_n &= \left(1 + 2R_n + \kappa_g\sqrt{\frac{t}{2n}}\right) \\ &\quad \left(2\sqrt{\frac{\tau t}{n}}(1 + R_n) + \sqrt{\frac{8L^2\sigma^2d_2}{\eta^2\zeta n} + 2\epsilon_n}\right), \\ \epsilon_n &= \frac{\hbar\tau t + 1 + R_n}{n},\end{aligned}$$

it holds that $\tilde{\rho}_n = O(\sqrt{d_2/n})$, $\epsilon_n = O(\sqrt{d_2}/n)$, and with probability at least $1 - (\lceil\log_2(\sqrt{LB\tau tn})\rceil + 1)e^{-t}$, for every $\theta \in \Theta$,

$$\begin{aligned}\mathbb{E}_{\mathbb{P}_{\text{true}}}[f_\theta] &\leq \mathbb{E}_{\mathbb{P}_n}[f_\theta] + \tilde{\rho}_n\mathbb{E}_{\mathbb{P}_n} \\ &\quad [l'(W_2\phi(W_1x), y)^2\|W_2\phi'(W_1x)\|_2^2]^{\frac{1}{2}} + \epsilon_n. \quad \clubsuit\end{aligned}$$

6. Concluding Remarks

In this paper, we developed finite-sample nonasymptotic performance guarantees for Wasserstein DRO and its associated variation regularization without suffering from the curse of dimensionality. These results help us to understand the empirical success of Wasserstein DRO and/or Lipschitz and gradient regularization. In the meantime, many issues worth investigating are left to future work.

More General Loss Function Families and Distribution Families. We restrict the families of loss functions consistent with Lemma 1 and Lemma 2 that establish the equivalence between Wasserstein DRO and variation regularization. One can extend the results to more general families such as nonsmooth losses using the results in Gao and Kleywegt (2017); see a follow-up work in An and Gao (2021). In Section 3, we adopt the widely used transportation inequalities T_p, $p \in [1, 2]$, which covers most sub-Gaussian distributions and works for loss functions of linear and quadratic growth. One may obtain results for more general distributions and loss functions by considering other families of transportation-information inequalities (Bolley and Villani 2005, Gozlan and Léonard 2007, Villani 2008, Gozlan and Léonard 2010). We remark that the finite-sample performance guarantee in Esfahani and Kuhn (2017), although suffering from the curse of dimensionality, does not restrict the loss function family other than the growth condition and works for any distribution that admits Wasserstein concentration.

Wasserstein Distance of Other Orders. We focus primarily on the case $p \in [1, 2]$. Indeed, our proof of Theorem 1 relies crucially on this setting where the tensorization lemma (Lemma EC.2 in Online Appendix EC.2.1) applies. We leave the study for other orders of Wasserstein distance, including another important case $p = \infty$ that has been widely considered in adversarial robust learning (Goodfellow et al. 2015, Shaham et al. 2018, Yin et al. 2019) to future work.

Complexity Theory Based on Variation of the Loss. We developed a local Rademacher complexity theory based on the variation of the loss. Investigation of these techniques in the context of other problems in statistical learning theory seems interesting, and hopefully would yield new results.

In summary, we hope our results can inspire more fruitful findings for problems in operations research and machine learning in which Wasserstein distributional robustness plays an increasingly prominent role.

References

Abdullah MA, Ren H, Ammar HB, Milenkovic V, Luo R, Zhang M, Wang J (2019) Wasserstein robust reinforcement learning. Preprint, submitted July 30, https://arxiv.org/abs/1907.13196.

An Y, Gao R (2021) Generalization bounds for (Wasserstein) robust optimization. *Adv. Neural Inform. Processing Systems* 34: 10382–10392.

Ban GY, Rudin C (2019) The big data newsvendor: Practical insights from machine learning. *Oper. Res.* 67(1):90–108.

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

Bansal N, Chen X, Wang Z (2018) Can we gain more from orthogonality regularizations in training deep networks? *Adv. Neural Inform. Processing Systems* 31:4261–4271.

Bartl D, Drapeau S, Obłój J, Wiesel J (2021) Sensitivity analysis of Wasserstein distributionally robust optimization problems. *Proc. Royal Soc. London A* 477(2256):20210176.

Bartlett PL, Mendelson S (2002) Rademacher and Gaussian complexities: Risk bounds and structural results. *J. Machine Learn. Res.* 3(Nov):463–482.

Bartlett PL, Bousquet O, Mendelson S (2005) Local rademacher complexities. *Ann. Statist.* 33(4):1497–1537.

Bayraksan G, Love DK (2015) Data-driven stochastic programming using phi-divergences. *The Operations Research Revolution* (INFORMS), 1–19.

Ben-Tal A, Den Hertog D, De Waegenaere A, Melenberg B, Rennen G (2013) Robust solutions of optimization problems affected by uncertain probabilities. *Management Sci.* 59(2):341–357.

Blanchet J, Kang Y (2020) Semi-supervised learning based on distributionally robust optimization. *Data Anal. Appl. 3: Comput. Classification, Financial Statist. Stochastic Methods* 5:1–33.

Blanchet J, Kang Y (2021) Sample out-of-sample inference based on Wasserstein distance. *Oper. Res.* 69(3):985–1013.

Blanchet J, Murthy K (2019) Quantifying distributional model risk via optimal transport. *Math. Oper. Res.* 44(2):565–600.

Blanchet J, Chen L, Zhou XY (2022) Distributionally robust mean-variance portfolio selection with Wasserstein distances. *Management Sci.* 68(9):6382–6410.

Blanchet J, Kang Y, Murthy K (2019a) Robust Wasserstein profile inference and applications to machine learning. *J. Appl. Probability* 56(3):830–857.

Blanchet J, Murthy K, Si N (2022a) Confidence regions in Wasserstein distributionally robust estimation. *Biometrika* 109(2): 295–315.

Blanchet J, Murthy K, Zhang F (2022b) Optimal transport-based distributionally robust optimization: Structural properties and iterative schemes. *Math. Oper. Res.* 47(2):1500–1529.

Blanchet J, Glynn PW, Yan J, Zhou Z (2019b) Multivariate distributionally robust convex regression under absolute error loss. *Adv. Neural Inform. Processing Systems* 32:11794–11803.

Bobkov SG, Götze F (1999) Exponential integrability and transportation cost related to logarithmic Sobolev inequalities. *J. Functional Anal.* 163(1):1–28.

Bolley F, Villani C (2005) Weighted csiszár-kullback-pinsker inequalities and applications to transportation inequalities. *Ann. Faculty Sci. Toulouse Math.* 14:331–352.

Bousquet O, Chapelle O, Hein M (2003) Measure based regularization. *Adv. Neural Inform. Processing Systems* 16:1221–1228.

Calafiore GC, El Ghaoui L (2006) On distributionally robust chance-constrained linear programs. *J. Optim. Theory Appl.* 130(1):1–22.

Carlsson JG, Behroozi M, Mihic K (2018) Wasserstein distance and the distributionally robust TSP. *Oper. Res.* 66(6):1603–1624.

Chen Y, Li W (2020) Optimal transport natural gradient for statistical manifolds with continuous sample space. *Inform. Geometry* 3(1): 1–32.

Chen R, Paschalidis IC (2018) A robust learning approach for regression models based on distributionally robust optimization. *J. Machine Learn. Res.* 19(1):517–564.

Chen H-B, Chewi S, Niles-Weed J (2021) Dimension-free log-sobolev inequalities for mixture distributions. *J. Funct. Anal.* 281(11): 109236.

Chen Z, Kuhn D, Wiesemann W (2023) On approximations of data-driven chance constrained programs over Wasserstein Balls. *Oper. Res.* Forthcoming.

Cherukuri A, Cortés J (2019) Cooperative data-driven distributionally robust optimization. *IEEE Trans. Automated Control* 65(10): 4400–4407.

Delage E, Ye Y (2010) Distributionally robust optimization under moment uncertainty with application to data-driven problems. *Oper. Res.* 58(3):595–612.

Derman E, Mannor S (2020) Distributional robustness and regularization in reinforcement learning. Preprint, submitted March 5, https://arxiv.org/abs/2003.02894.

Djellout H, Guillin A, Wu L (2004) Transportation cost-information inequalities and applications to random dynamical systems and diffusions. *Ann. Probability* 32(3B):2702–2732.

Duan C, Fang W, Jiang L, Yao L, Liu J (2018) Distributionally robust chance-constrained approximate AC-OPF with Wasserstein metric. *IEEE Trans. Power Systems* 33(5):4924–4936.

Duchi JC, Namkoong H (2019) Variance-based regularization with convex objectives. *J. Machine Learn. Res.* 20(68):1–55.

Duchi JC, Glynn PW, Namkoong H (2021) Statistics of robust optimization: A generalized empirical likelihood approach. *Math. Oper. Res.* 46(3):946–969.

Duchi J, Hashimoto T, Namkoong H (2020) Distributionally robustlosses for latent covariate mixtures. Preprint, submitted July 28, https://arxiv.org/abs/2007.13982.

Erdoğan E, Iyengar G (2006) Ambiguous chance constrained problems and robust optimization. *Math. Programming* 107(1-2):37–61.

Esfahani PM, Kuhn D (2018) Data-driven distributionally robust optimization using the Wasserstein metric: Performance guarantees and tractable reformulations. *Math. Programming* 171(1): 115–166.

Finlay C, Calder J, Abbasi B, Oberman A (2018) Lipschitz regularized deep neural networks generalize and are adversarially robust. Preprint, submitted August 28, https://arxiv.org/abs/1808.09540.

Fournier N, Guillin A (2015) On the rate of convergence in Wasserstein distance of the empirical measure. *Probability Theory Related Fields* 162(3-4):707–738.

Gao R, Kleywegt AJ (2022) Distributionally robust stochastic optimization with Wasserstein distance. *Math. Oper. Res.* 48(2):603–655.

Gao R, Kleywegt AJ (2017) Distributionally robust stochastic optimization with dependence structure. Preprint, submitted January 16, https://arxiv.org/abs/1701.04200.

Gao R, Xie L, Xie Y, Xu H (2018) Robust hypothesis testing using Wasserstein uncertainty sets. *Adv. Neural Inform. Processing Systems* 31:7902–7912.

Goh J, Sim M (2010) Distributionally robust optimization and its tractable approximations. *Oper. Res.* 58(4-part-1):902–917.

Goodfellow I, Shlens J, Szegedy C (2015) Explaining and harnessing adversarial examples. *Proc. Internat. Conf. on Learn. Representations* (ICLR, San Diego), https://doi.org/10.48550/arXiv.1412.6572.

Gozlan N, Léonard C (2007) A large deviation approach to some transportation cost inequalities. *Probability Theory Related Fields* 139(1-2):235–283.

Gozlan N, Léonard C (2010) Transport inequalities. A survey. *Markov Processing Related Fields* 16:635–736.

Gulrajani I, Ahmed F, Arjovsky M, Dumoulin V, Courville AC (2017) Improved training of Wasserstein GANs. *Adv. Neural Inform. Processing Systems* 30:5767–5777.

Huang L, Liu X, Lang B, Yu AW, Wang Y, Li B (2018) Orthogonal weight normalization: Solution to optimization over multiple dependent stiefel manifolds in deep neural networks. *Proc. 32nd AAAI Conf. on Artificial Intelligence* (AAAI, Palo Alto, CA), 3271–3278.

Jakubovitz D, Giryes R (2018) Improving DNN robustness to adversarial attacks using Jacobian regularization. *Adv. Neural Inform. Processing Systems* 30:5767–5777.

Jiang R, Guan Y (2016) Data-driven chance constrained stochastic program. *Math. Programming* 158(1):291–327.

Karimi H, Nutini J, Schmidt M (2016) Linear convergence of gradient and proximal-gradient methods under the Polyak-Łojasiewicz

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

condition. *Proc. Joint Eur. Conf. on Machine Learn. and Knowledge Discovery in Databases* (Springer, Berlin), 795–811.

Koltchinskii V (2006) Local Rademacher complexities and oracle inequalities in risk minimization. *Ann. Statist.* 34(6):2593–2656.

Kuhn D, Esfahani PM, Nguyen VA, Shafieezadeh-Abadeh S (2019) Wasserstein distributionally robust optimization: Theory and applications in machine learning. *Operations Research & Management Science in the Age of Analytics* (INFORMS), 130–166.

Lam H (2019) Recovering best statistical guarantees via the empirical divergence-based distributionally robust optimization. *Oper. Res.* 67(4):1090–1105.

Lee J, Raginsky M (2018) Minimax statistical learning with Wasserstein distances. *Adv. Neural Inform. Processing Systems* 31:2687–2696.

Levine A, Feizi S (2020) Wasserstein smoothing: Certified robustness against Wasserstein adversarial attacks. *Proc. Internat. Conf. on Artificial Intelligence and Statist.*, PMLR 108:3938–3947.

Li W, Montúfar G (2018) Natural gradient via optimal transport. *Inform. Geometry* 1(2):181–214.

Lopez AT, Jog V (2018) Generalization error bounds using Wasserstein distances. *Proc. IEEE Inform. Theory Workshop*, 1–5.

Luo F, Mehrotra S (2019) Decomposition algorithm for distributionally robust optimization using Wasserstein metric with an application to a class of regression models. *Eur. J. Oper. Res.* 278(1):20–35.

Luxburg Uv, Bousquet O (2004) Distance-based classification with Lipschitz functions. *J. Machine Learn. Res.* 5(Jun):669–695.

Lyu C, Huang K, Liang HN (2015) A unified gradient regularization family for adversarial examples. *Proc. IEEE Internat. Conf. on Data Mining*(IEEE, Piscataway, NJ), 301–309.

Miyato T, Kataoka T, Koyama M, Yoshida Y (2018) Spectral normalization for generative adversarial networks. *Proc. Internat. Conf. on Learn. Representations* (ICLR, Vancouver, BC, Canada), https://openreview.net/forum?id=B1QRgziT-.

Najafi A, Maeda SI, Koyama M, Miyato T (2019) Robustness to adversarial perturbations in learning from incomplete data. *Advances in Neural Information Processing Systems* (Curran Associates, Inc., Redhook, NY), 5542–5552.

Nguyen VA, Kuhn D, Mohajerin Esfahani P (2021a) Distributionally robust inverse covariance estimation: The Wasserstein shrinkage estimator. *Oper. Res.* 70(1):490–515.

Nguyen VA, Shafieezadeh-Abadeh S, Kuhn D, Mohajerin Esfahani P (2021b) Bridging bayesian and minimax mean square error estimation via Wasserstein distributionally robust optimization. *Math. Oper. Res.* 48(1):1–37.

Oberman AM, Calder J (2018) Lipschitz regularized deep neural networks converge and generalize. Preprint, submitted August 28, https://arxiv.org/abs/1808.09540.

Popescu I (2007) Robust mean-covariance solutions for stochastic optimization. *Oper. Res.* 55(1):98–112.

Russo D, Zou J (2019) How much does your data exploration overfit? Controlling bias via information usage. *IEEE Trans. Inform. Theory* 66(1):302–323.

Scarf H (1958) A min-max solution of an inventory problem. *Studies in the Mathematical Theory of Inventory and Production* (Stanford University Press, Stanford, CA).

Shafieezadeh-Abadeh S, Esfahani PM, Kuhn D (2015) Distributionally robust logistic regression. *Adv. Neural Inform. Processing Systems* 1:1576–1584.

Shafieezadeh-Abadeh S, Kuhn D, Esfahani PM (2019) Regularization via mass transportation. *J. Machine Learn. Res.* 20(103): 1–68.

Shafieezadeh-Abadeh S, Nguyen VA, Kuhn D, Esfahani PM (2018) Wasserstein distributionally robust Kalman filtering. *Proc. 32nd Internat. Conf. on Neural Inform. Processing Systems* (Curran Associates Inc., Red Hook, NY), 8483–8492.

Shaham U, Yamada Y, Negahban S (2018) Understanding adversarial training: Increasing local stability of supervised models through robust optimization. *Neurocomput.* 307:195–204.

Shalev-Shwartz S, Ben-David S (2014) *Understanding Machine Learning: From Theory to Algorithms* (Cambridge University Press, Cambridge, UK).

Shapiro A, Kleywegt A (2002) Minimax analysis of stochastic problems. *Optim. Methods Software* 17(3):523–542.

Shapiro A, Dentcheva D, Ruszczynski A (2014) *Lectures on Stochastic Programming: Modeling and Theory*, vol. 16 (SIAM, Philadelphia).

Sinha A, Namkoong H, Duchi J (2018) Certifying some distributional robustness with principled adversarial training. *Proc. Internat. Conf. on Learn. Representations*, https://openreview.net/forum?id=Hk6kPgZA-.

Smirnova E, Dohmatob E, Mary J (2019) Distributionally robust reinforcement learning. Preprint, submitted February 23, https://arxiv.org/abs/1902.08708.

Sokolić J, Giryes R, Sapiro G, Rodrigues MR (2017) Robust large margin deep neural networks. *IEEE Trans. Signal Processing* 65(16):4265–4280.

Terjék D (2019) Adversarial Lipschitz regularization. *Proc. Internat. Conf. on Learn. Representations*. https://openreview.net/forum?id=Bke_DertPB.

Villani C (2008) *Optimal Transport: Old and New*, vol. 338 (Springer Science & Business Media, New York).

Virmaux A, Scaman K (2018) Lipschitz regularity of deep neural networks: Analysis and efficient estimation. *Adv. Neural Inform. Processing Systems* 31:3835–3844.

Volpi R, Namkoong H, Sener O, Duchi JC, Murino V, Savarese S (2018) Generalizing to unseen domains via adversarial data augmentation. *Adv. Neural Inform. Processing Systems* 31:5334–5344.

Vorontsov E, Trabelsi C, Kadoury S, Pal C (2017) On orthogonality and learning recurrent networks with long term dependencies. *Proc. 34th Internat. Conf. on Machine Learn.*, PMLR 70:3570–3578.

Wainwright MJ (2019) *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, vol. 48 (Cambridge University Press, Cambridge, UK).

Wang Z, Glynn PW, Ye Y (2016) Likelihood robust optimization for data-driven problems. *Comput. Management Sci.* 13(2): 241–261.

Wang H, Diaz M, Santos Filho JCS, Calmon FP (2019) An information-theoretic view of generalization via Wasserstein distance. *Proc. IEEE Internat. Sympos. on Inform. Theory* (IEEE, Piscataway, NJ), 577–581.

Wang H, Huang Y, Gao R, Calmon F (2021) Analyzing the generalization capability of sgld using properties of Gaussian channels. *Adv. Neural Inform. Processing Systems* 34:24222–24234.

Wang C, Gao R, Qiu F, Wang J, Xin L (2018a) Risk-based distributionally robust optimal power flow with dynamic line rating. *IEEE Trans. Power Systems* 33(6):6074–6086.

Wang C, Gao R, Wei W, Shafie-khah M, Bi T, Catalao JP (2018b) Risk-based distributionally robust optimal gas-power flow with Wasserstein distance. *IEEE Trans. Power Systems* 34(3): 2190–2204.

Wang Z, Glynn PW, Ye Y (2016) Likelihood robust optimization for data-driven problems. *Comput. Management Sci.* 13(2):241–261.

Wei C, Ma T (2019) Data-dependent sample complexity of deep neural networks via Lipschitz augmentation. *Adv. Neural Inform. Processing Systems* 32:9725–9736.

Wiesemann W, Kuhn D, Sim M (2014) Distributionally robust convex optimization. *Oper. Res.* 62(6):1358–1376.

Wozabal D (2014) Robustifying convex risk measures for linear portfolios: A nonparametric approach. *Oper. Res.* 62(6):1302–1315.

Xie W (2021) On distributionally robust chance constrained programs with Wasserstein distance. *Math. Programming* 186(1):115–155.

Downloaded from informs.org by [60.29.153.49] on 25 September 2024, at 01:59 . For personal use only, all rights reserved.

Xie D, Xiong J, Pu S (2017) All you need is beyond a good INIT: Exploring better solution for training extremely deep convolutional neural networks with orthonormality and modulation. *Proc. IEEE Conf. on Comput. Vision and Pattern Recognition* (IEEE Computer Society, Los Alamitos, CA), 5075–5084.

Xie W, Zhang J, Ahmed S (2021) Distributionally robust bottleneck combinatorial problems: Uncertainty quantification and robust decision making. *Math. Programming* 1–44.

Xu H, Mannor S (2012) Robustness and generalization. *Machine Learn.* 86(3):391–423.

Xu A, Raginsky M (2017) Information-theoretic analysis of generalization capability of learning algorithms. *Adv. Neural Inform. Processing Systems* 30:2524–2533.

Yang I (2017) A convex optimization approach to distributionally robust Markov decision processes with Wasserstein distance. *IEEE Control Systems Lett.* 1(1):164–169.

Yang I (2020) Wasserstein distributionally robust stochastic control: A data-driven approach. *IEEE Trans. Automated Control* 66(10): 3863–3870.

Yin D, Kannan R, Bartlett P (2019) Rademacher complexity for adversarially robust generalization. *Proc. Internat. Conf. on Machine Learn.*, PLMR 97:7085–7094.

Yu T, Thomas G, Yu L, Ermon S, Zou JY, Levine S, Finn C, Ma T (2020) Mopo: Model-based offline policy optimization. *Adv. Neural Inform. Processing Systems* 33:14129–14142.

Žáčková J (1966) On minimax solutions of stochastic linear programming problems. *Časopis Pěstování Matematiky* 91(4):423–430.

Zhao C, Guan Y (2018) Data-driven risk-averse stochastic optimization with Wasserstein metric. *Oper. Res. Lett.* 46(2):262–267.

Rui Gao is an assistant professor in the McCombs School of Business at the University of Texas at Austin. His main research studies data-driven decision making under uncertainty and prescriptive data analytics. His research has been recognized with several INFORMS paper competition awards including Winner in Junior Faculty Interest Group Paper Competition (2020) and Finalist in George Nicholson Student Paper Competition (2016).